

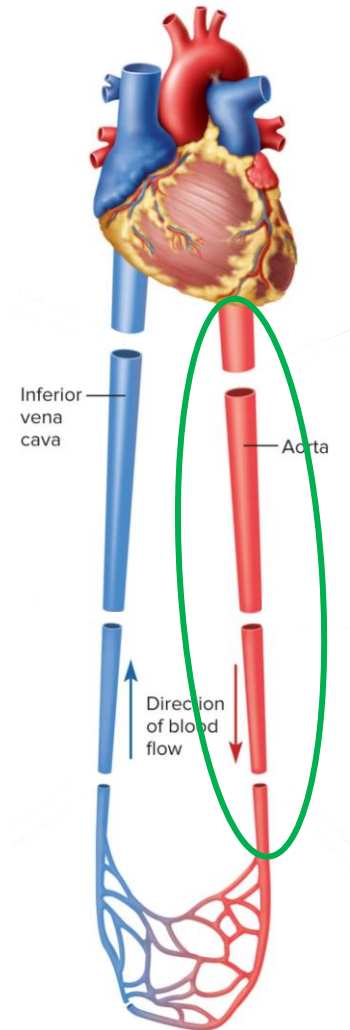
Exercise 15: Blood Vessels



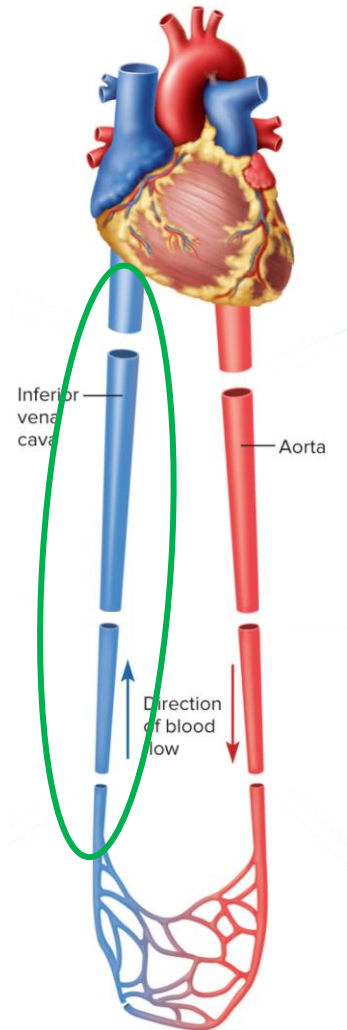
Blood Vessels

- *three principal categories of blood vessels

- ****arteries*** carry blood away from the heart



Blood Vessels

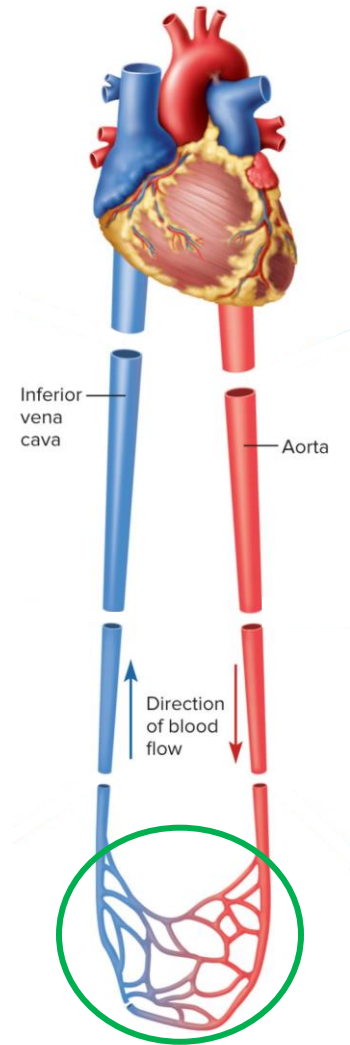


*three principal categories of blood vessels

***arteries** carry blood away from the heart

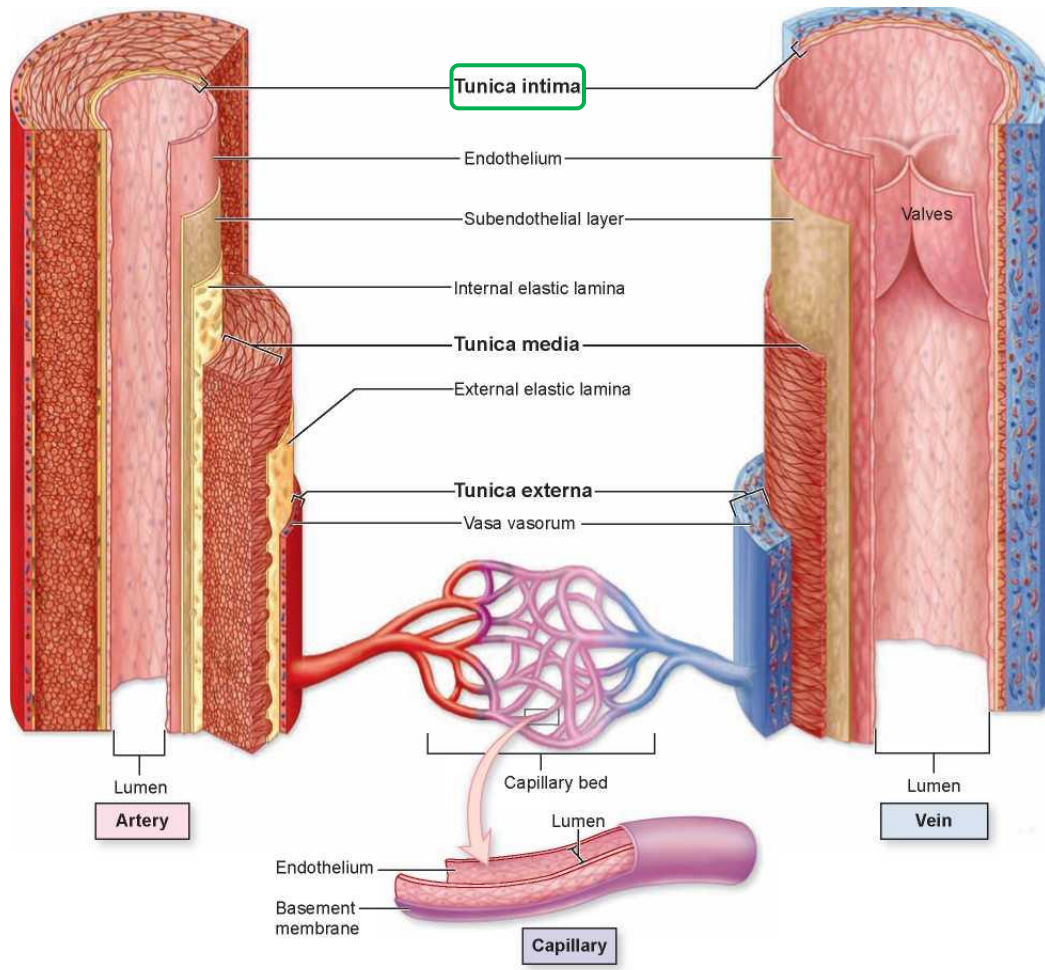
***veins** carry blood back to the heart

Blood Vessels



- *three principal categories of blood vessels
- ***arteries** carry blood away from the heart
- ***veins** carry blood back to the heart
- ***capillaries** are microscopic, thin-walled vessels that connect the smallest arteries to the smallest veins

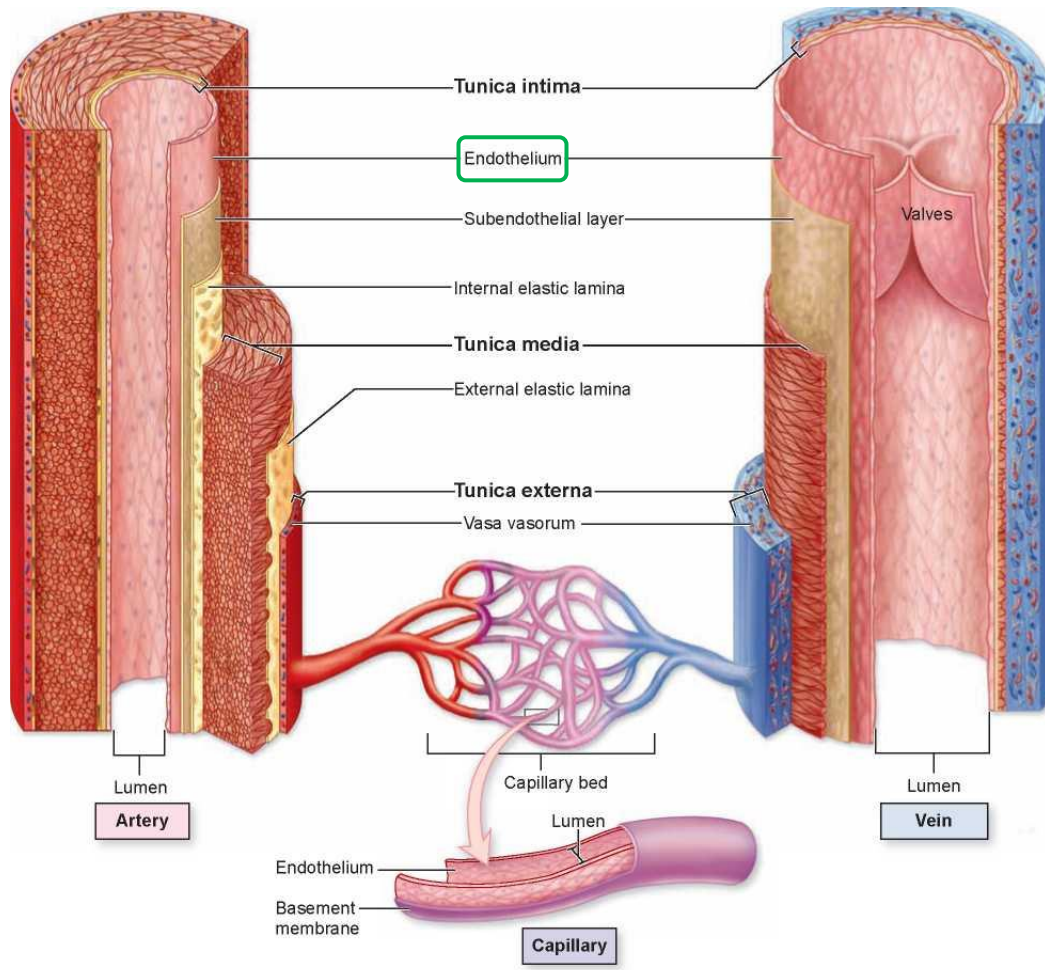
Blood Vessels



*walls of arteries and veins are composed of three layers called ***tunics***

****tunica interna (tunica intima)*** lines the inside of the vessel (exposed to blood)

Blood Vessels

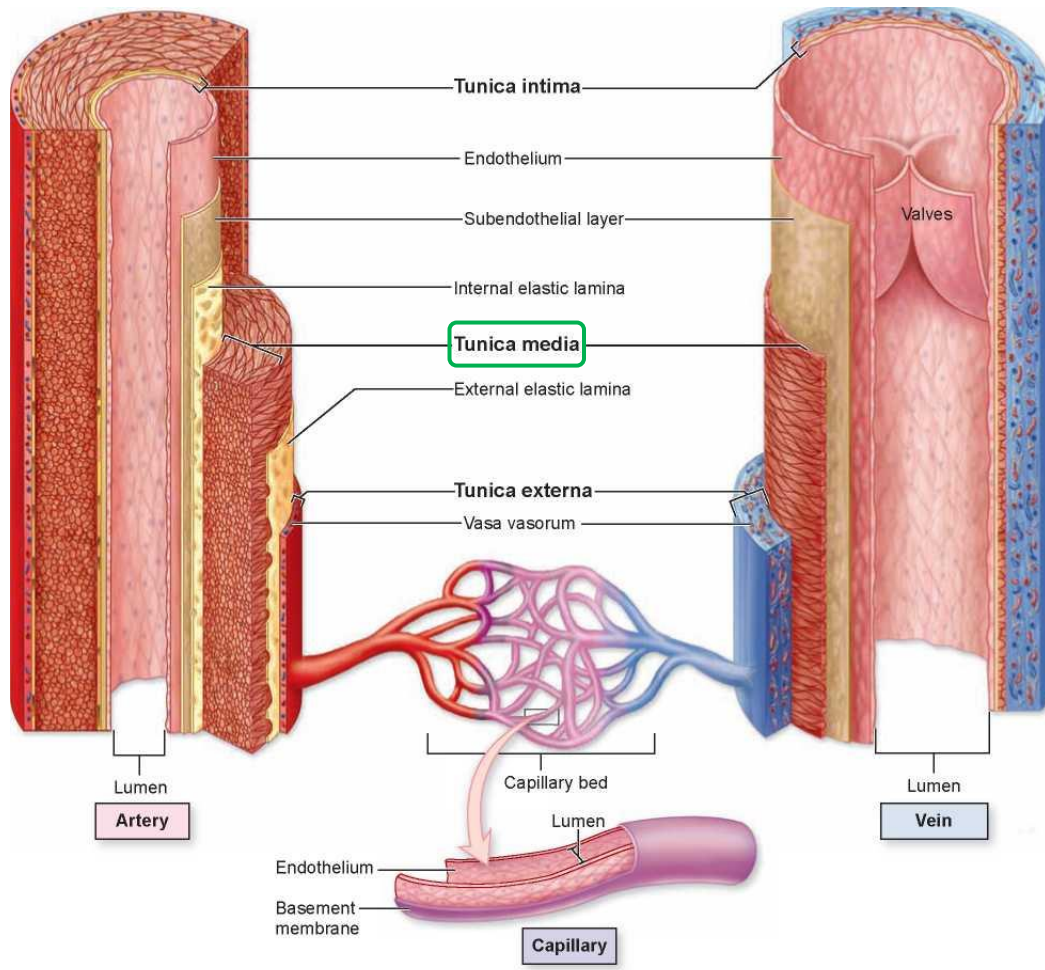


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Blood Vessels



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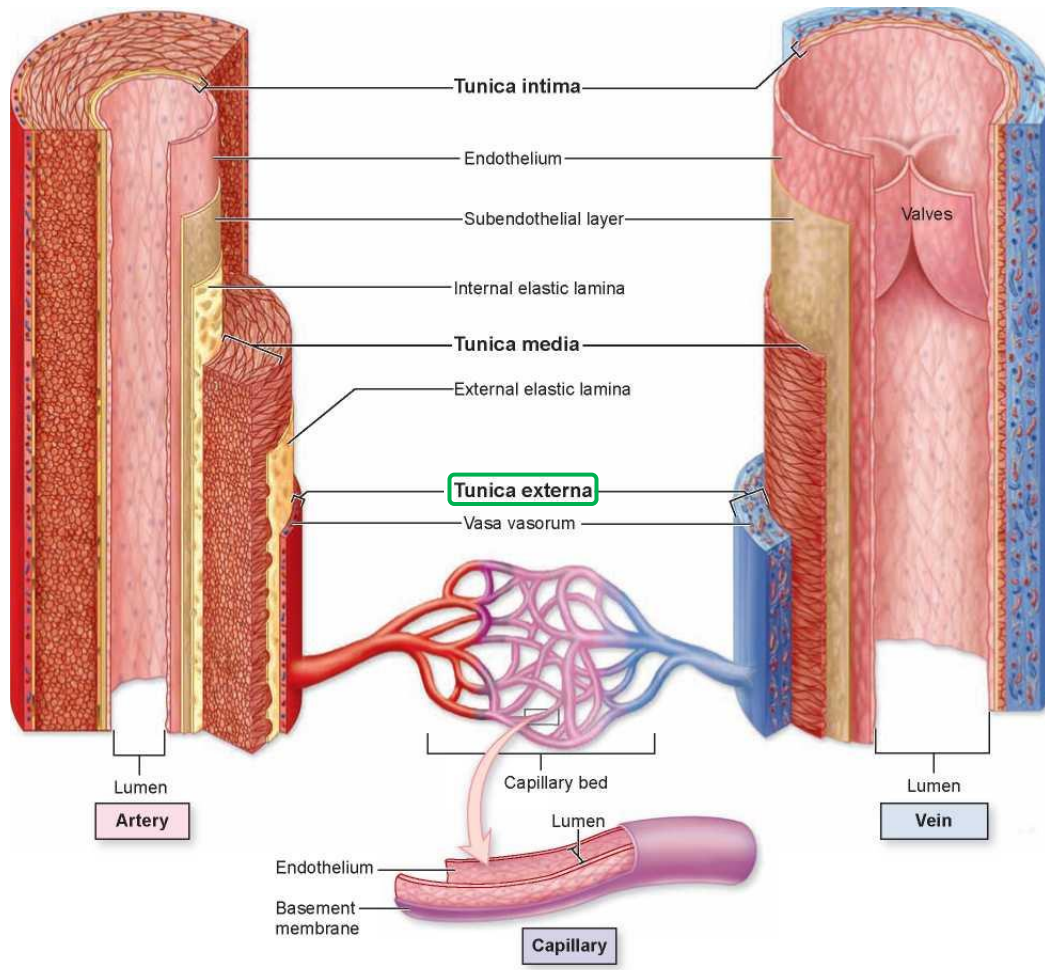
- ****tunica interna (tunica intima)*** lines the inside of the vessel (exposed to blood)

- *simple squamous epithelium (***endothelium***) that acts as a selectively permeable barrier

- ****tunica media*** is the middle and usually the thickest layer

- *consists of smooth muscle, collagen and sometimes elastic tissue

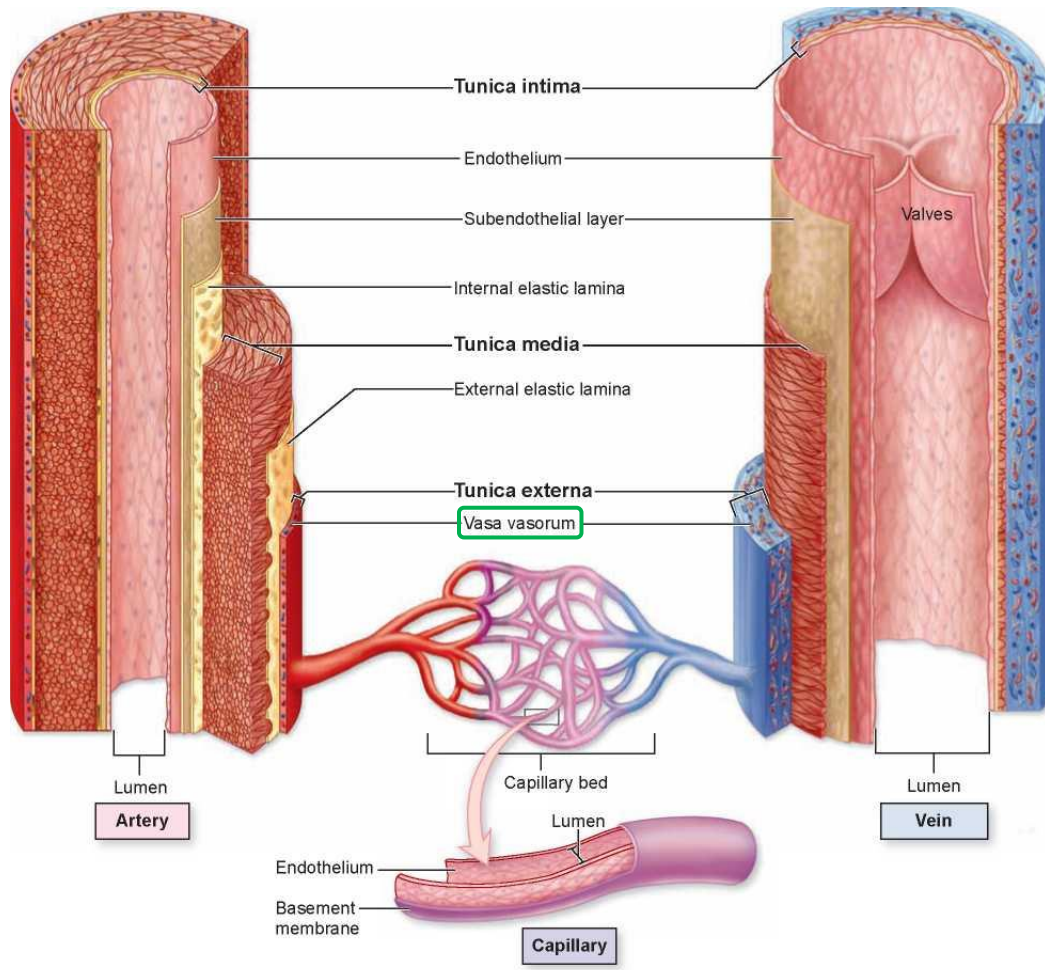
Blood Vessels



****tunica externa (tunica adventitia)*** is the outermost layer consisting of areolar connective tissue

*anchors the vessel to adjacent tissues and provides passage for small nerves, lymphatic vessels and smaller blood vessels

Blood Vessels

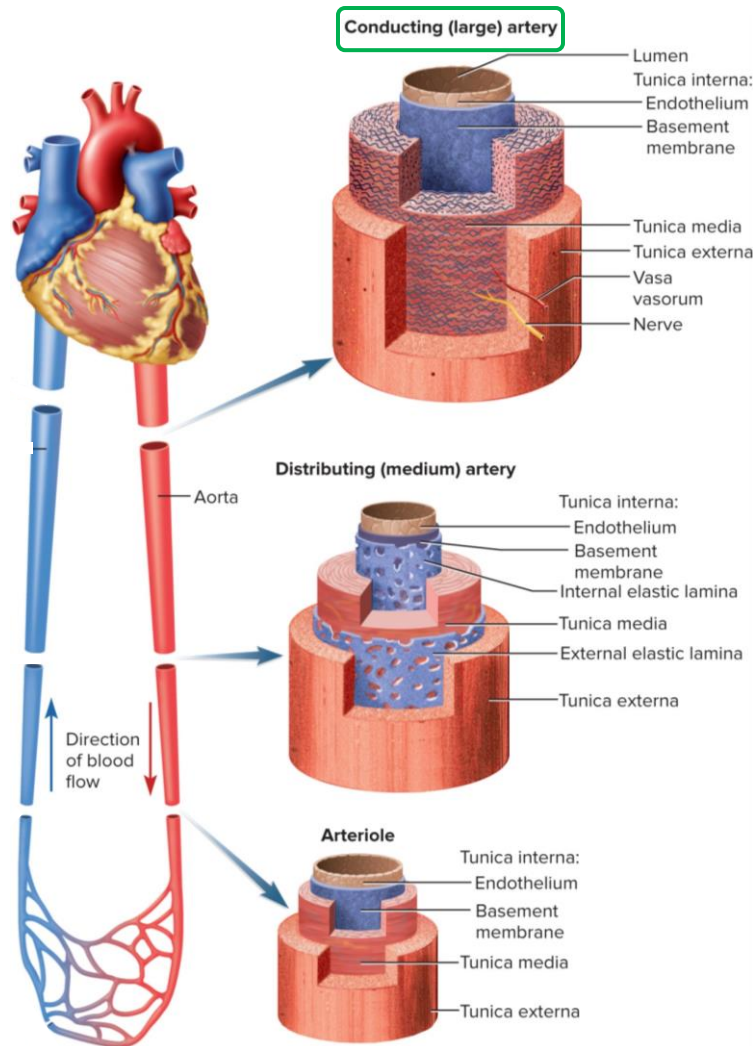


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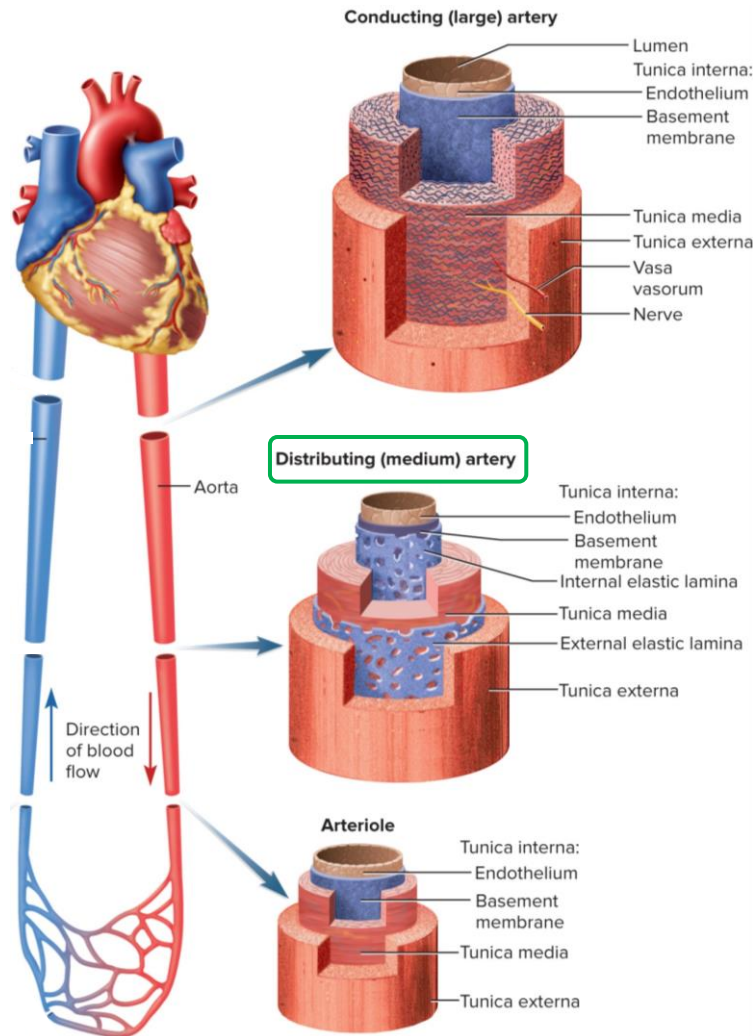
* ***vasa vasorum*** is a network of small vessels that nourish the deeper tissues of a larger blood vessel

Blood Vessels



****conducting (elastic or large) arteries*** are the biggest (aorta, common carotid and subclavian arteries, pulmonary trunk and common iliac arteries)

Blood Vessels

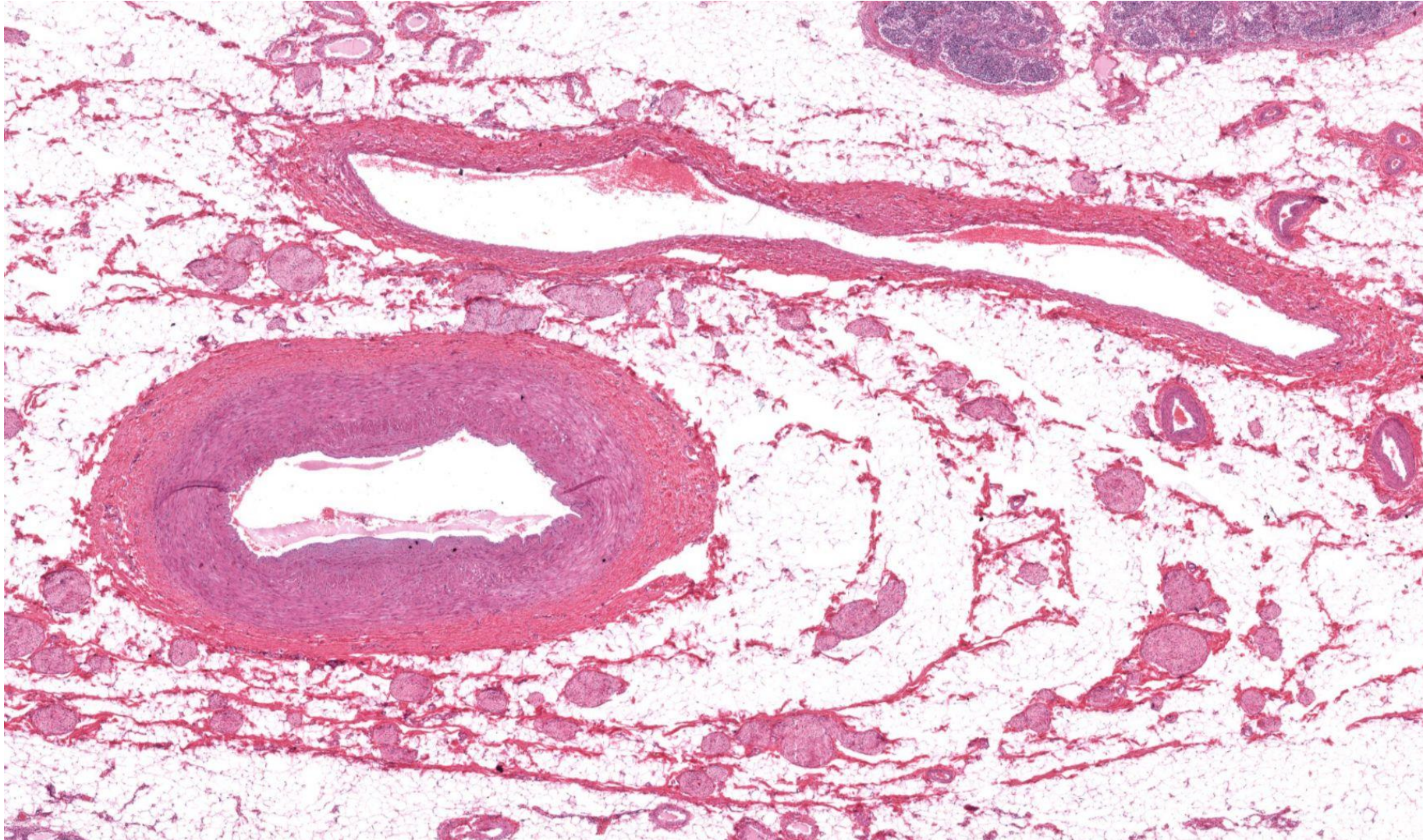


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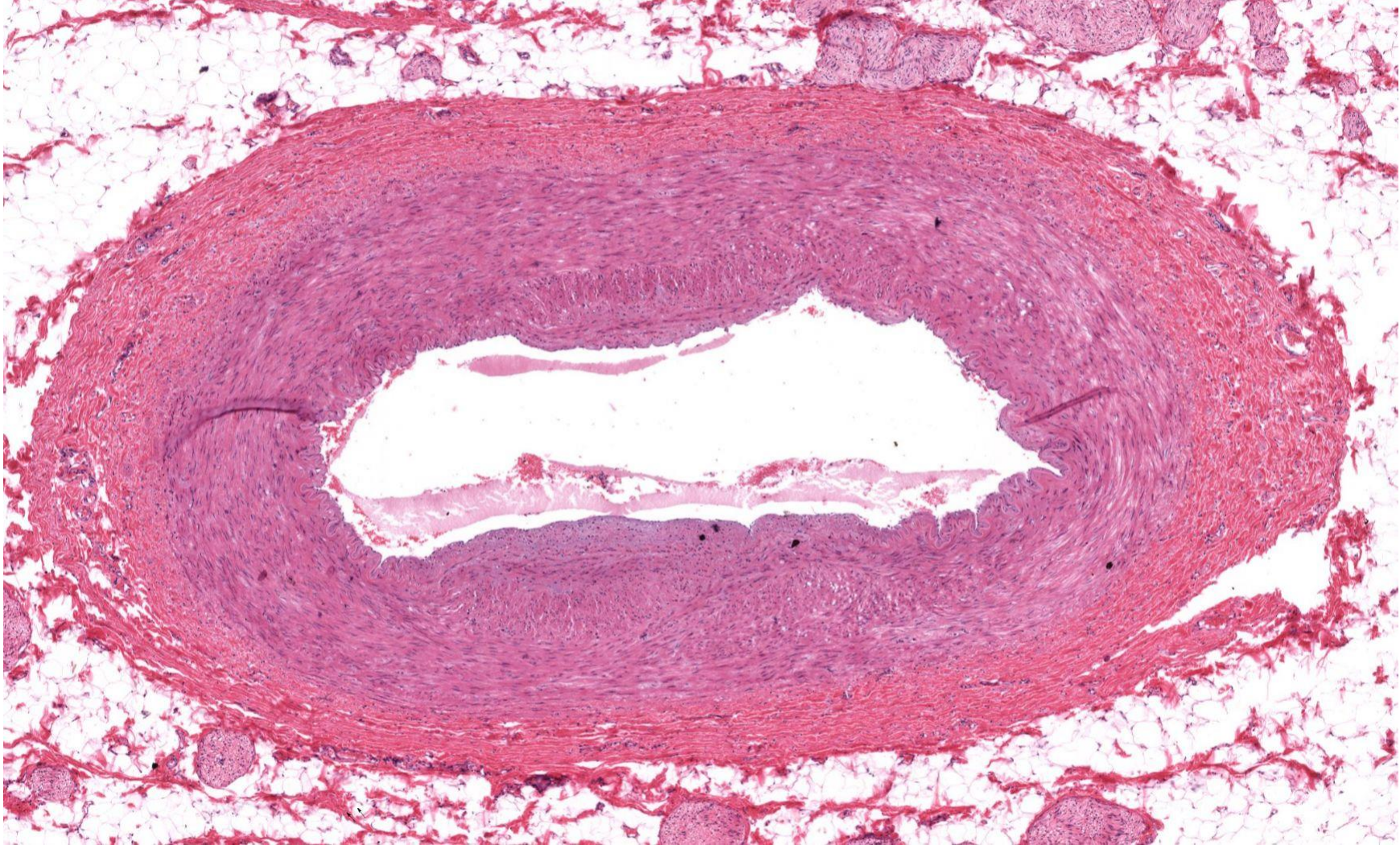
* ***distributing (muscular or medium) arteries*** are smaller branches that distribute blood to specific organs

* include brachial, femoral, and renal arteries

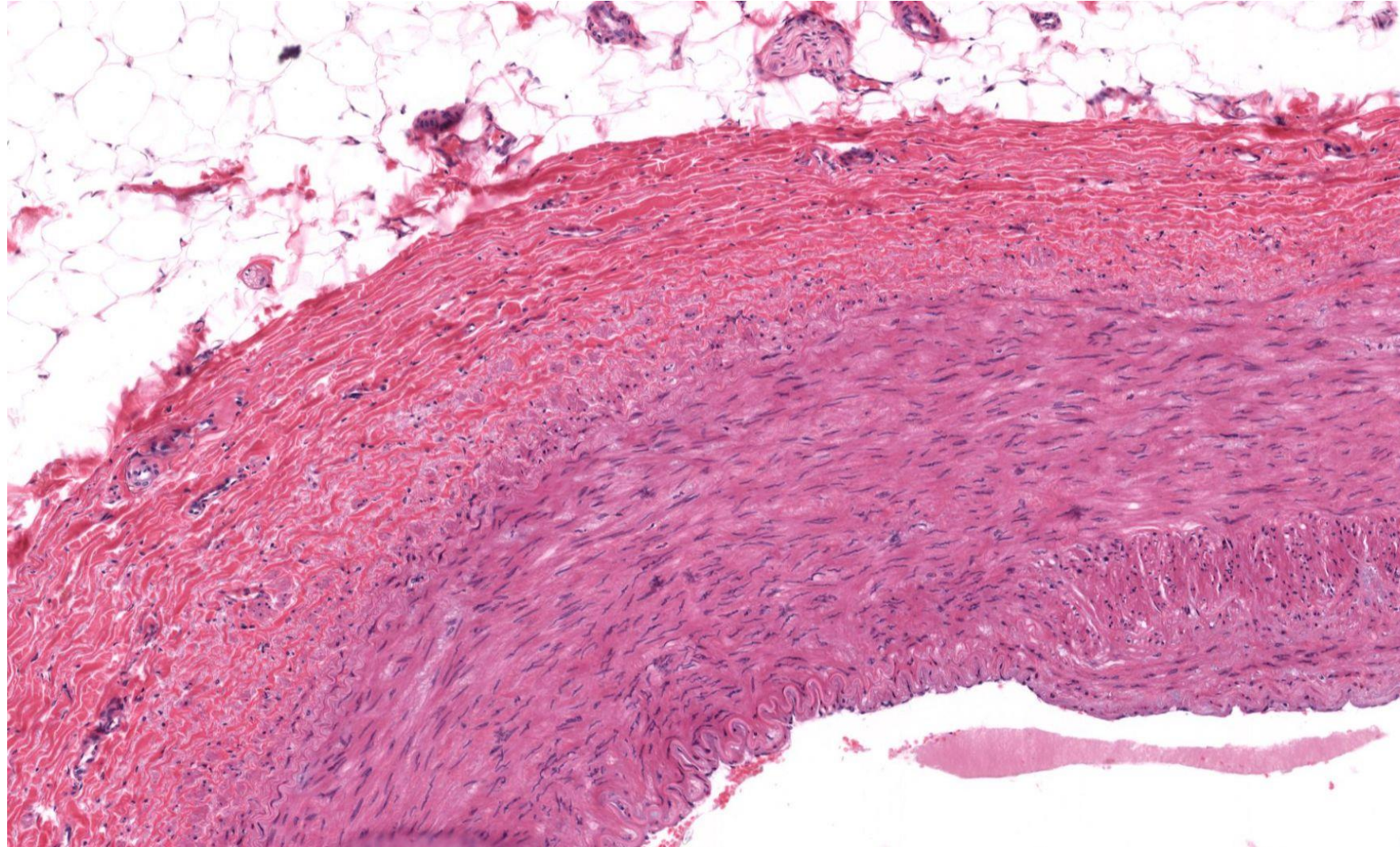
Histology



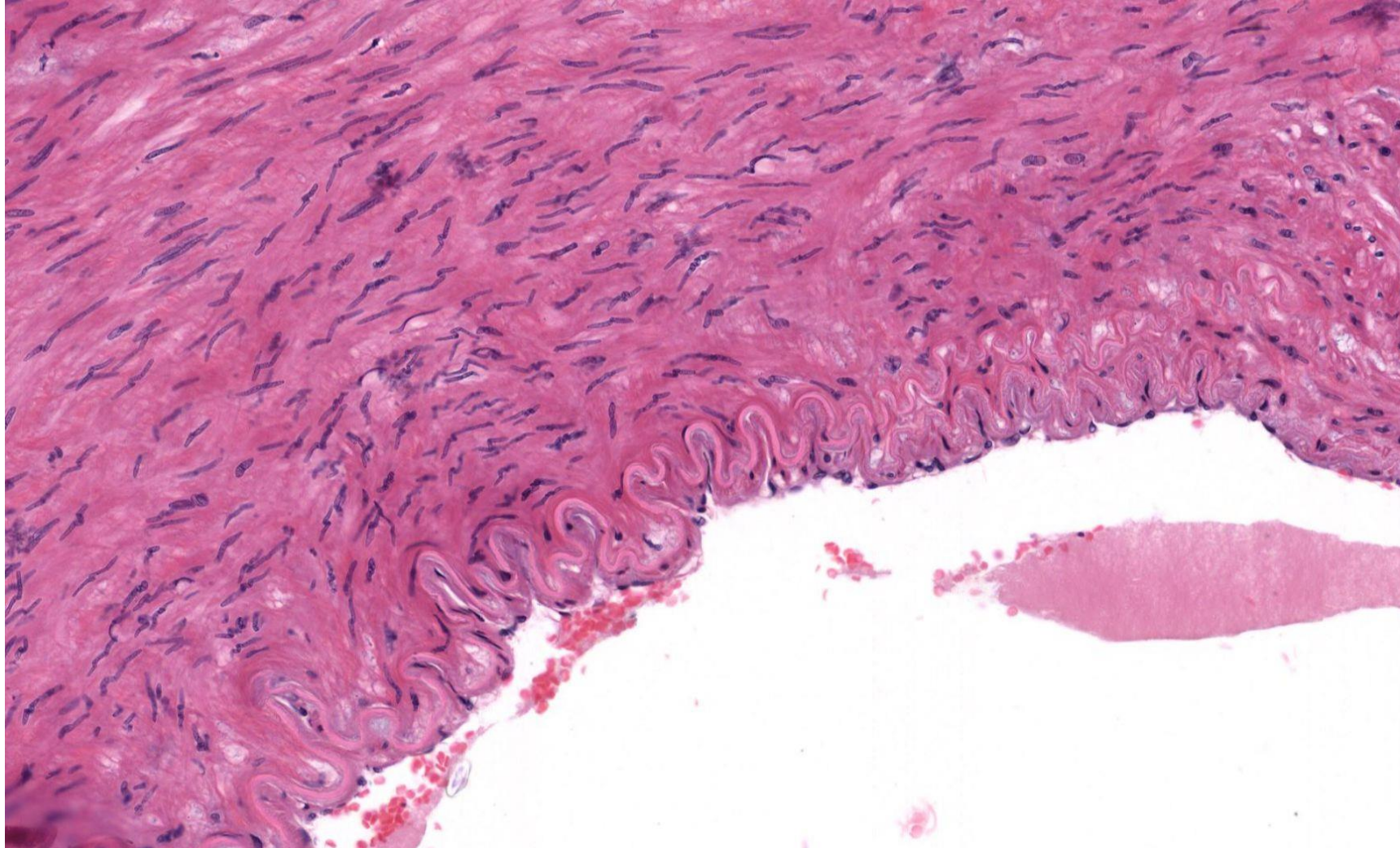
Histology



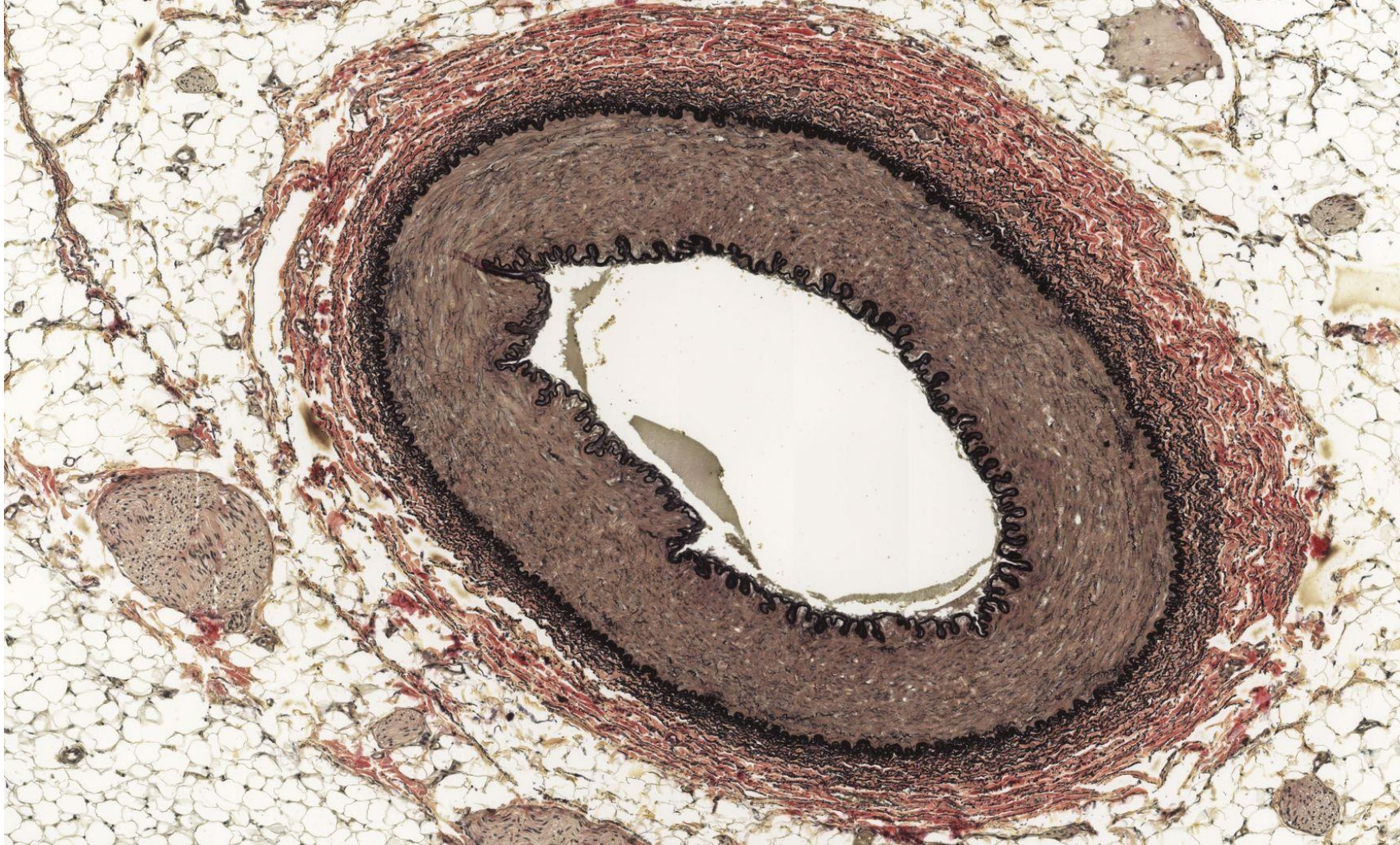
Histology

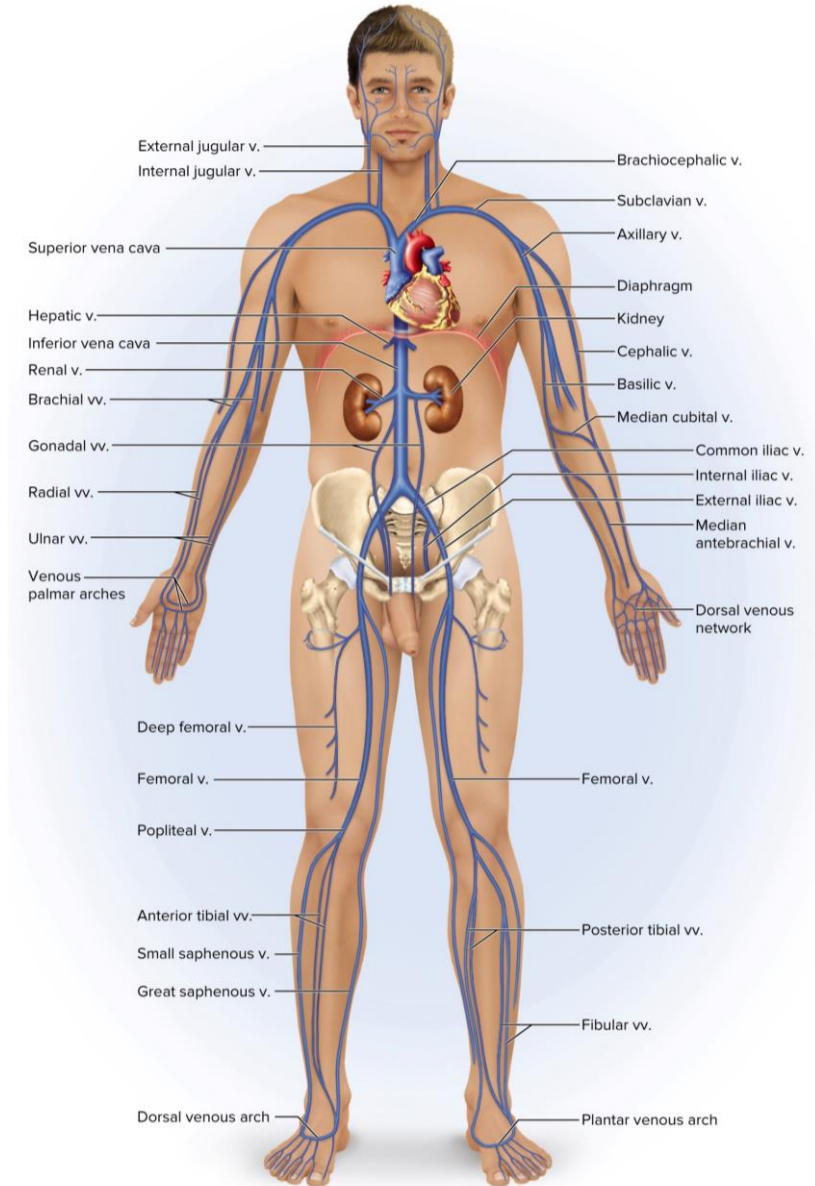
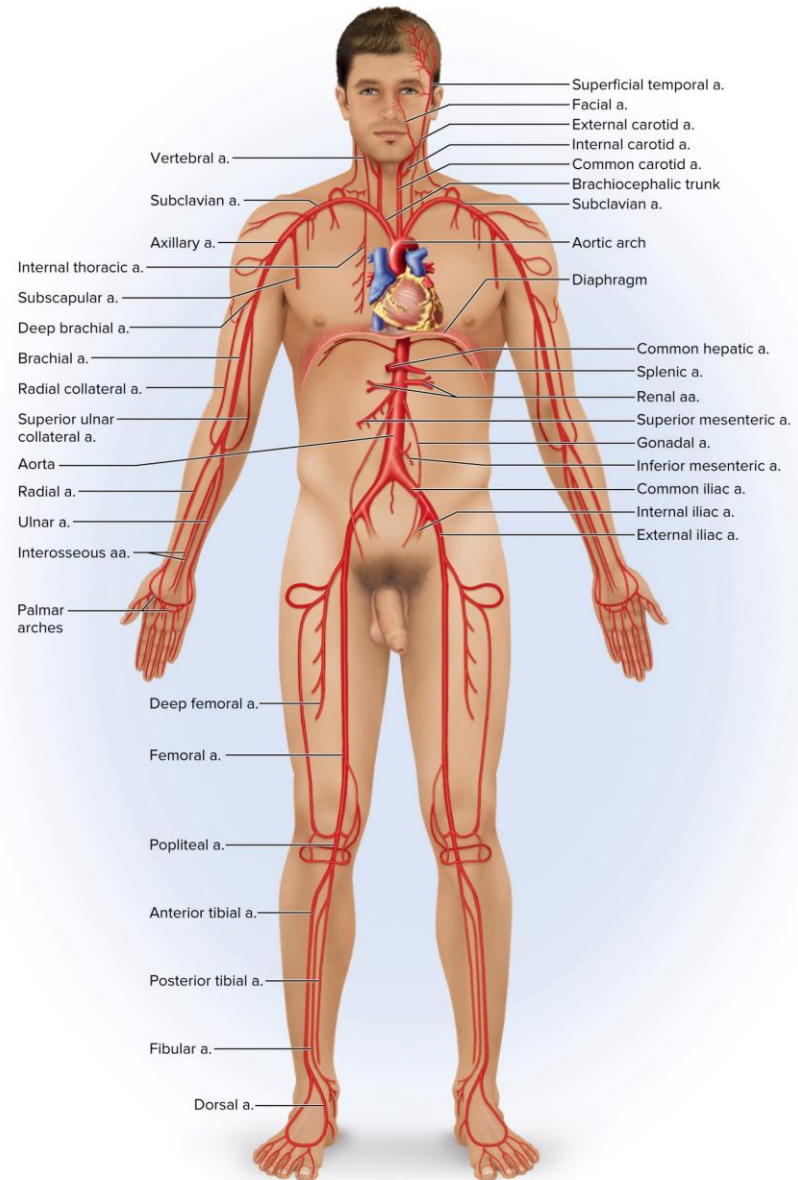


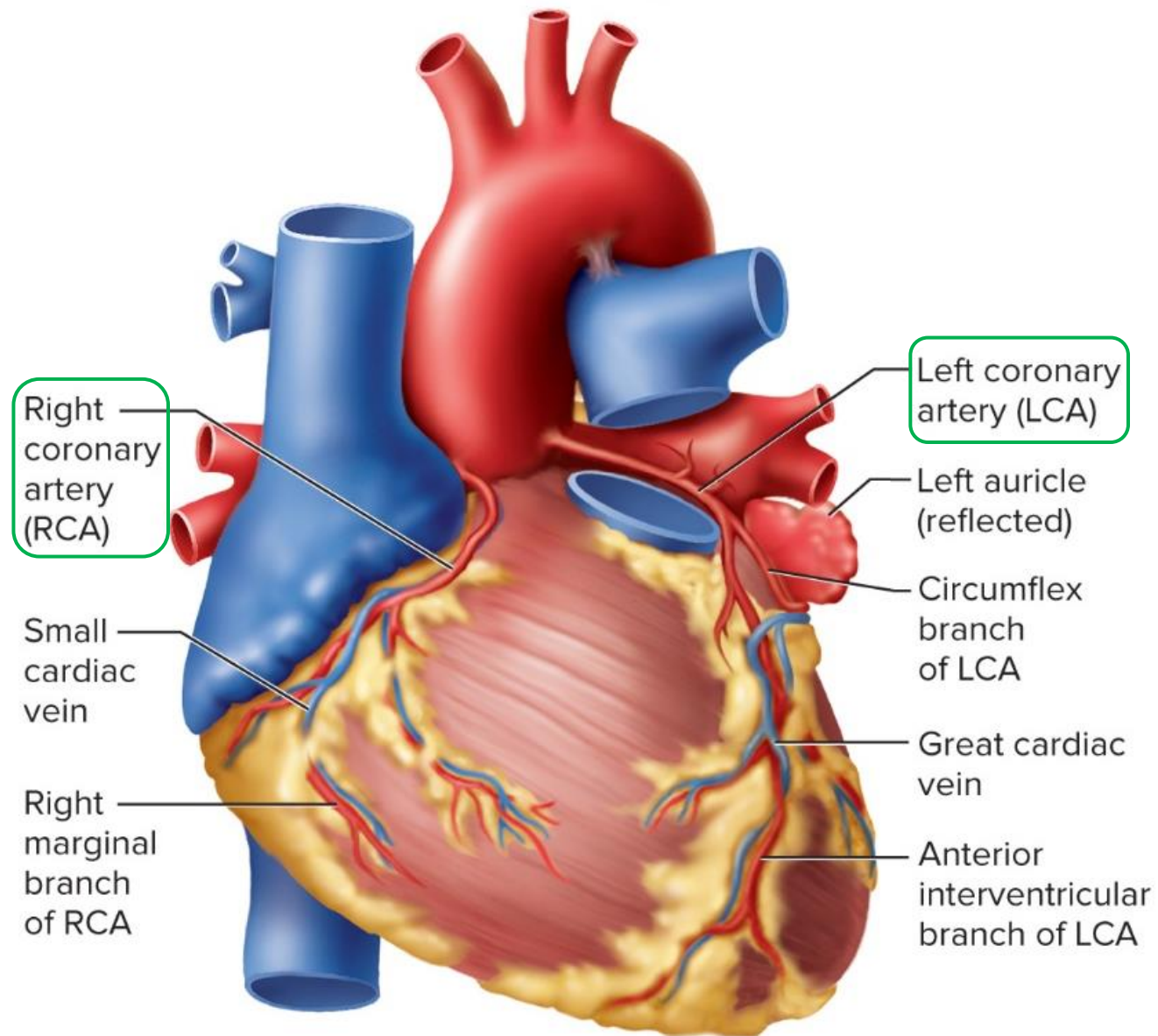
Histology

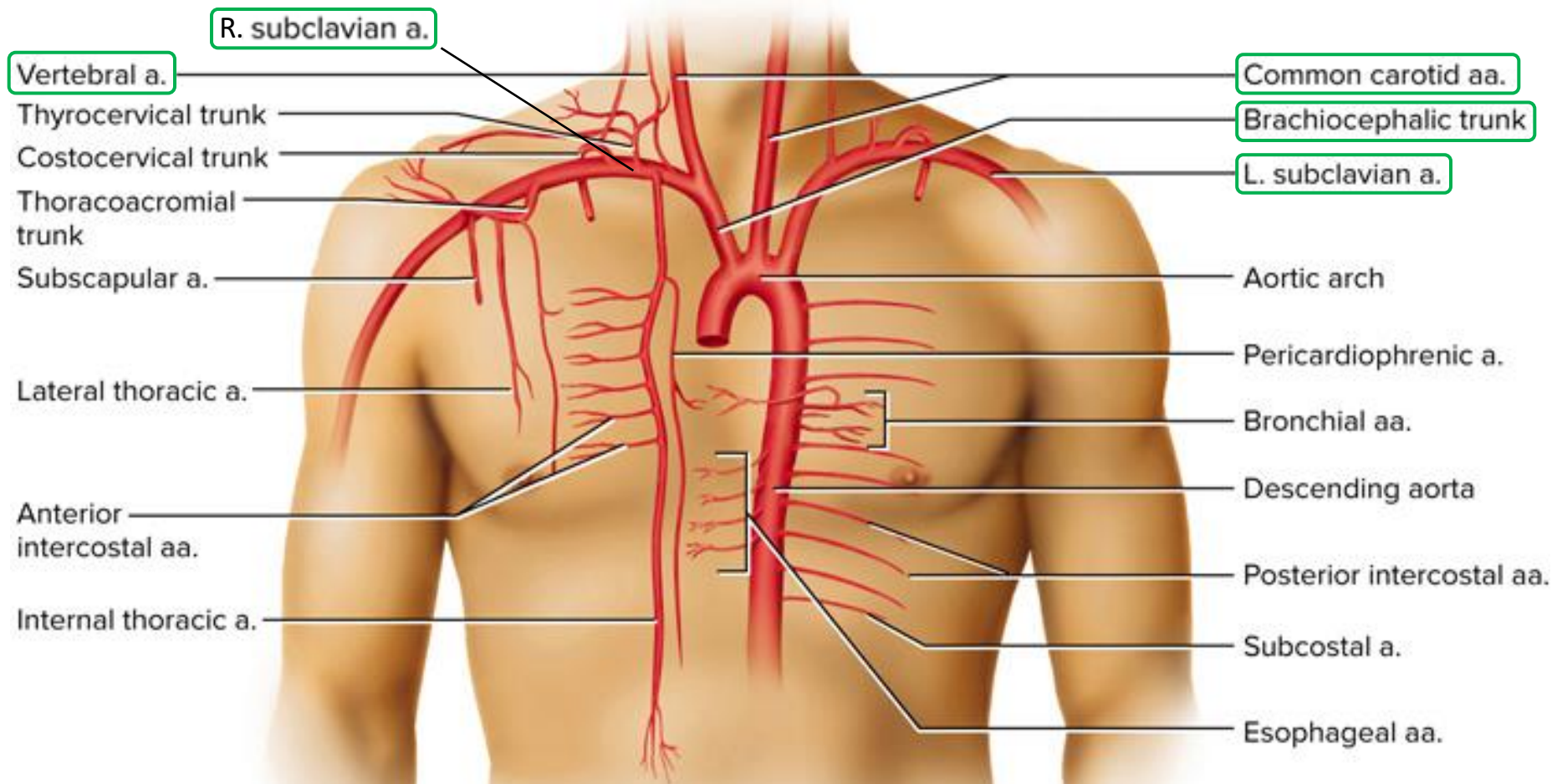


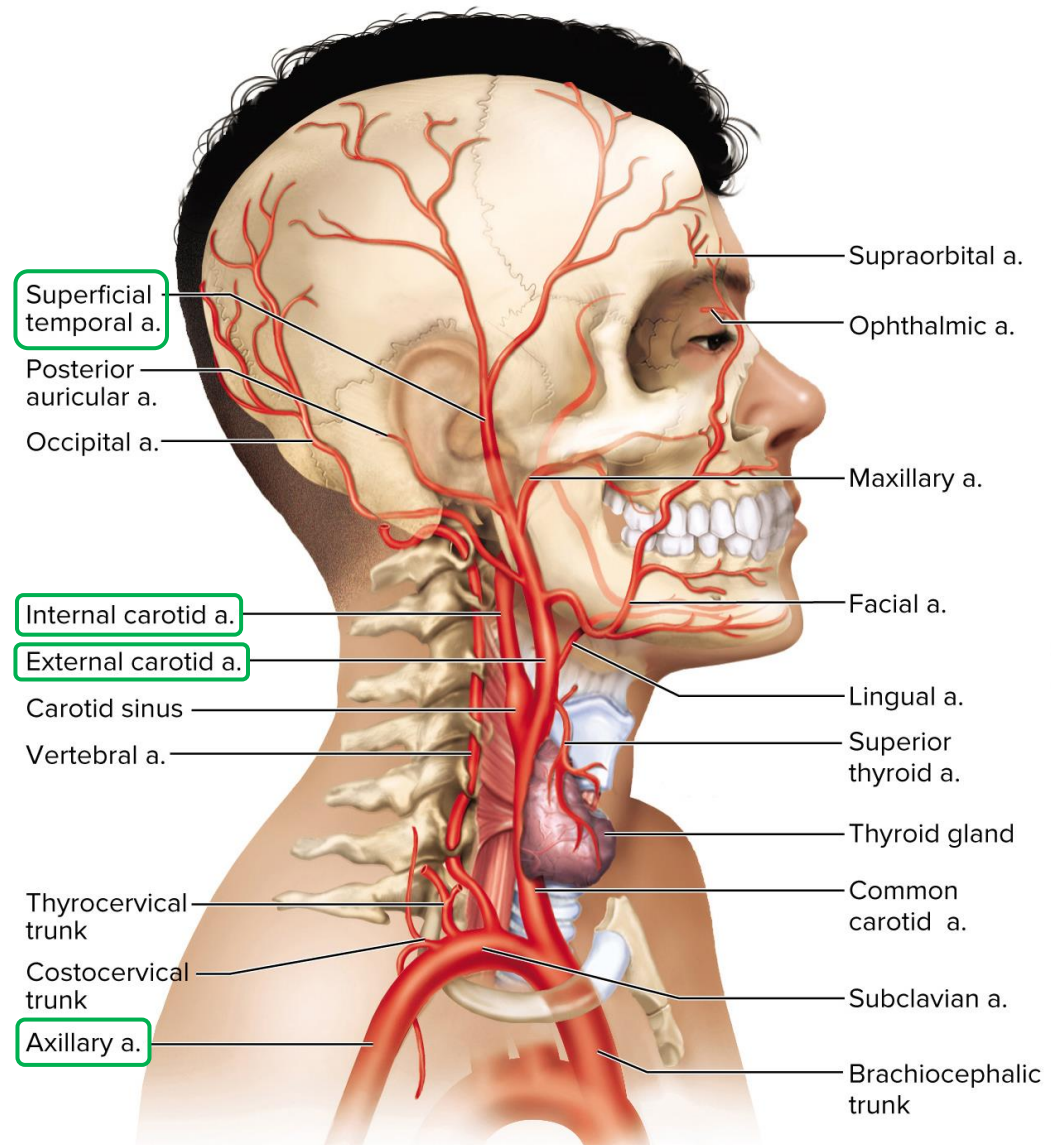
Histology

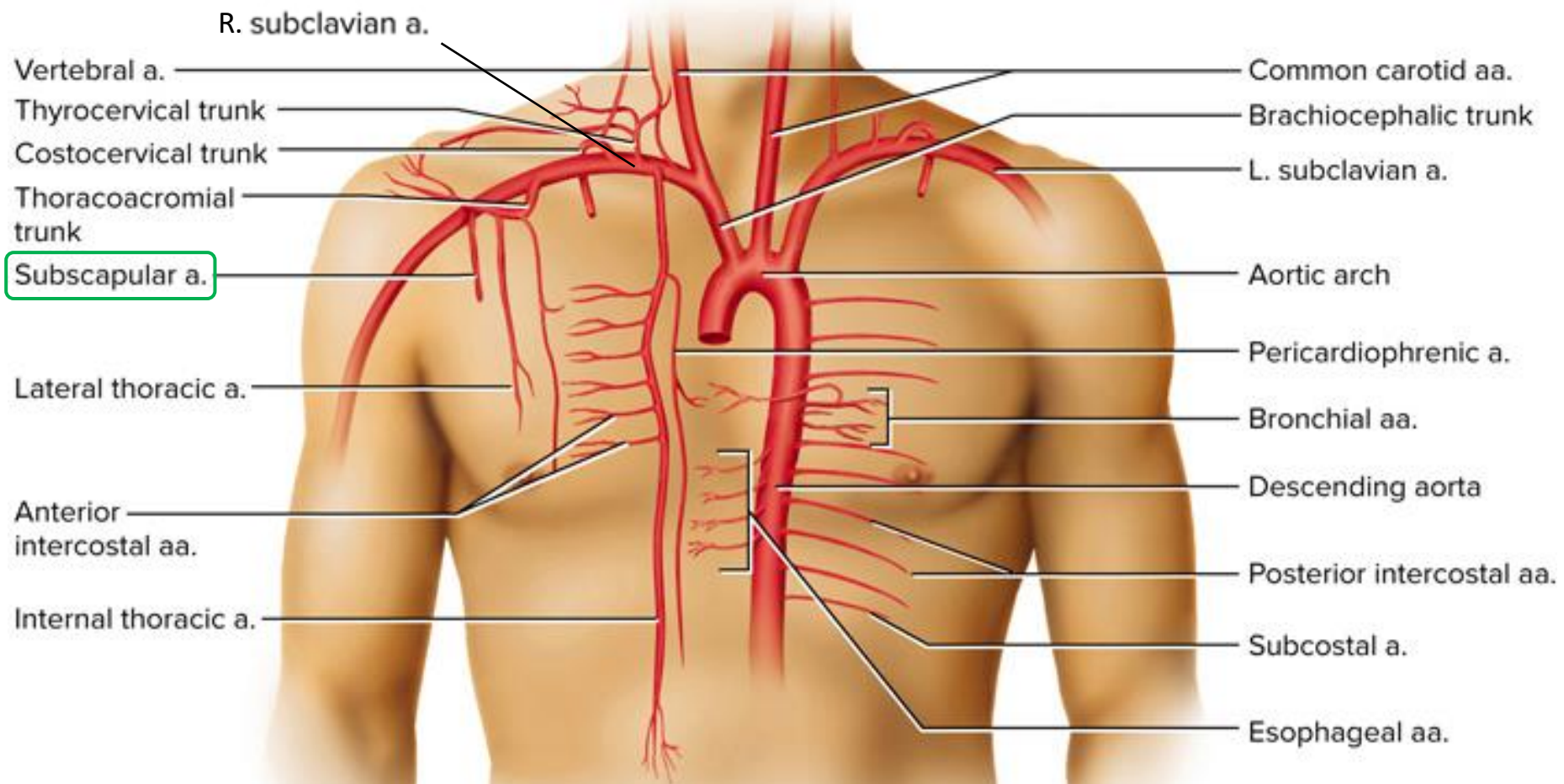


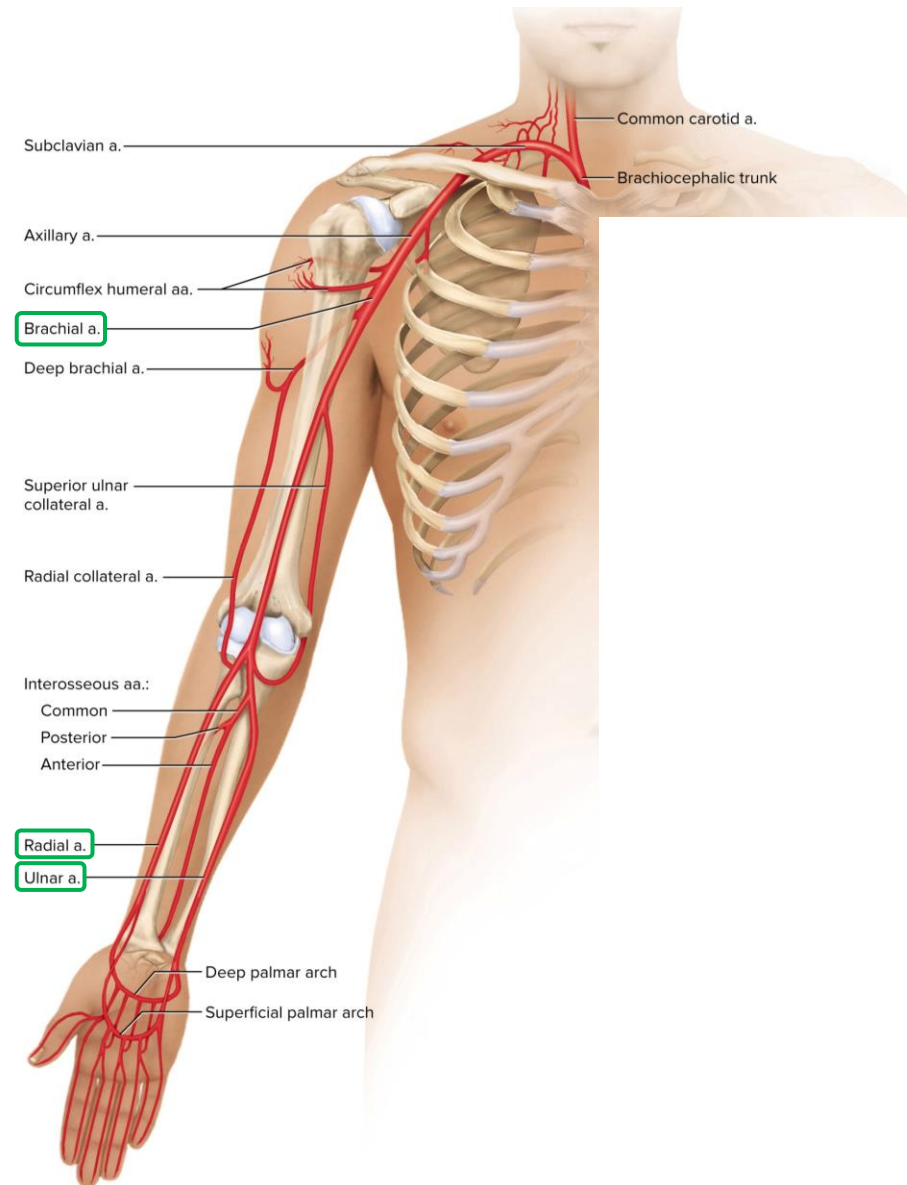


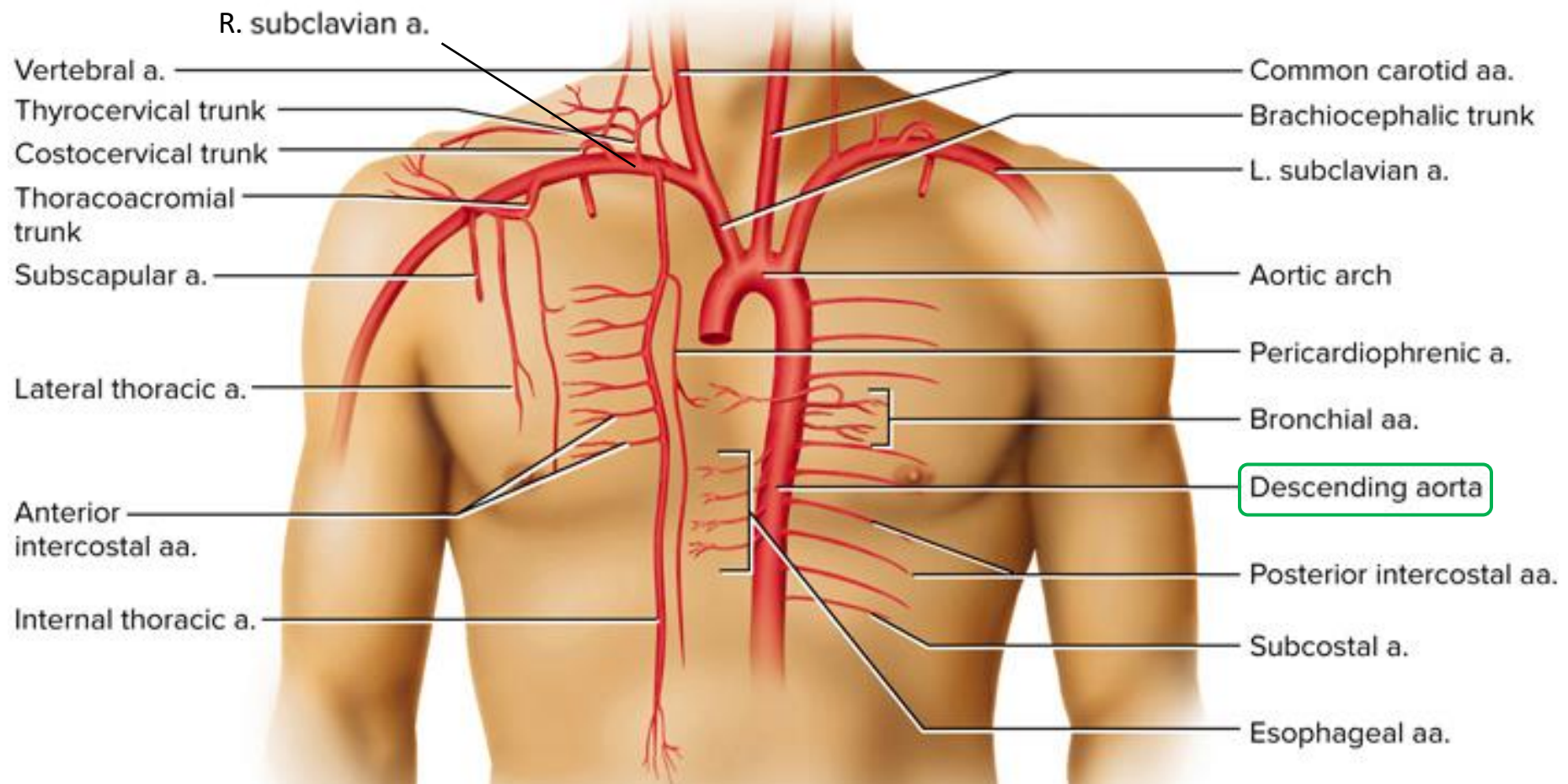


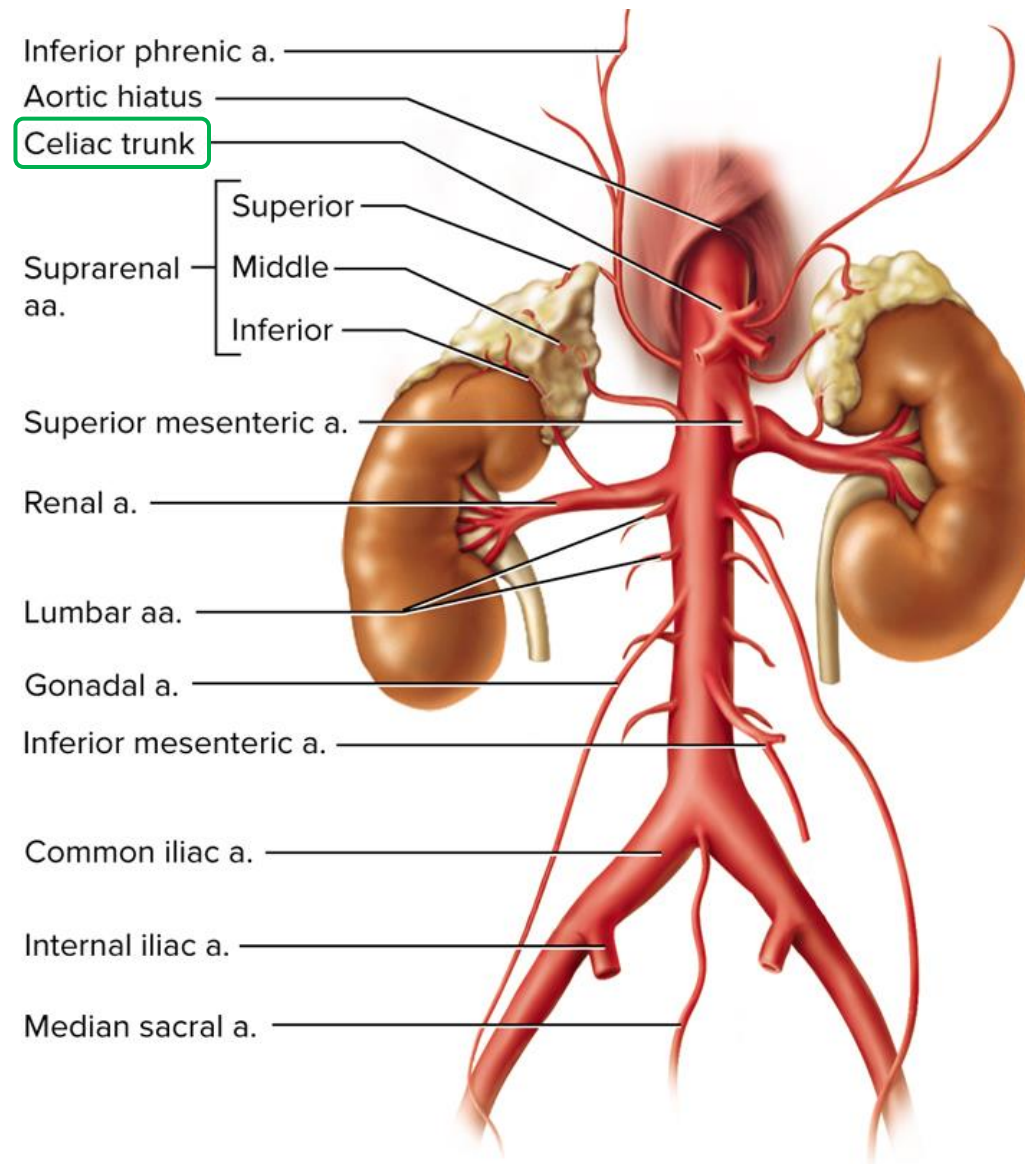


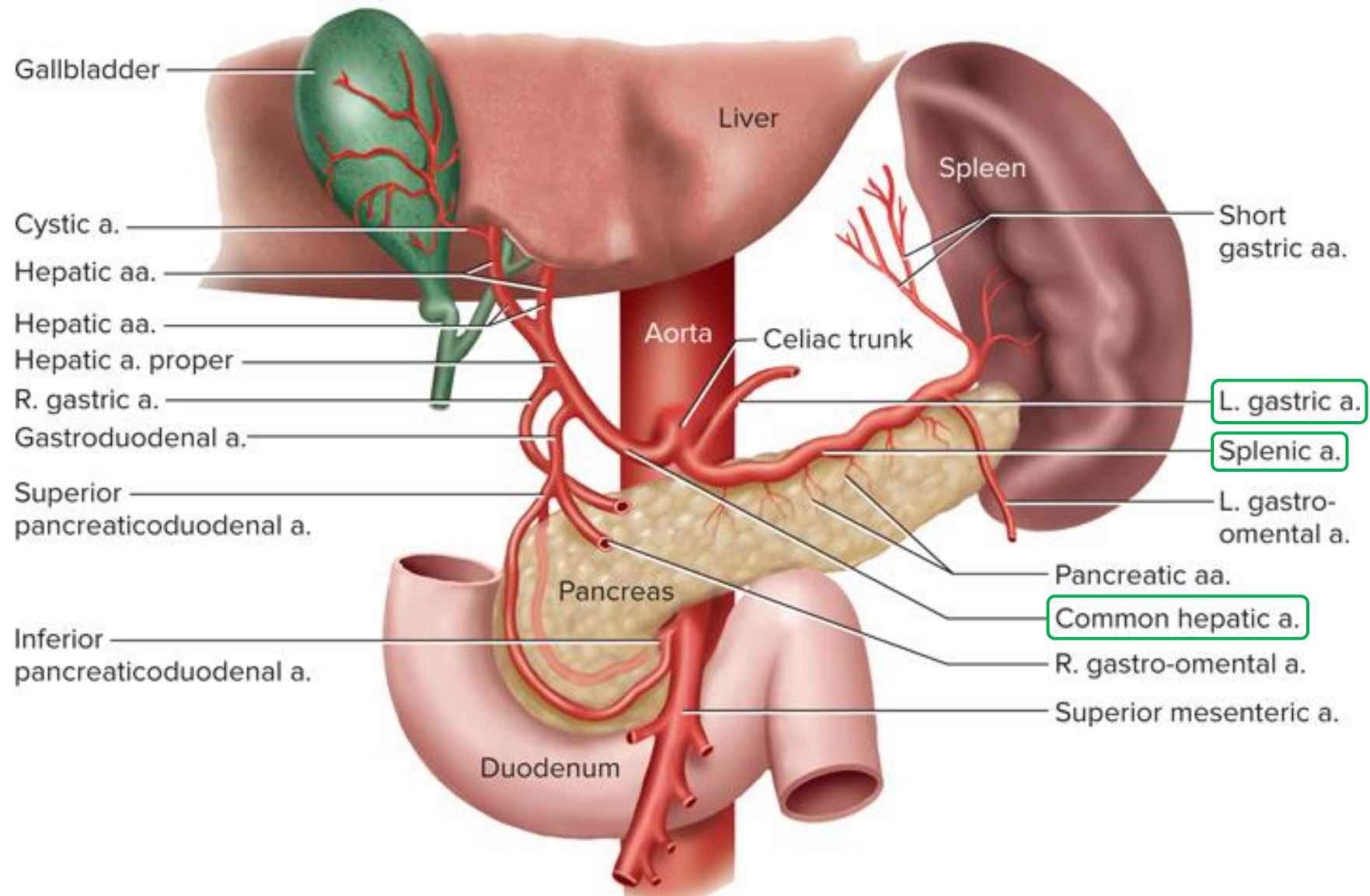


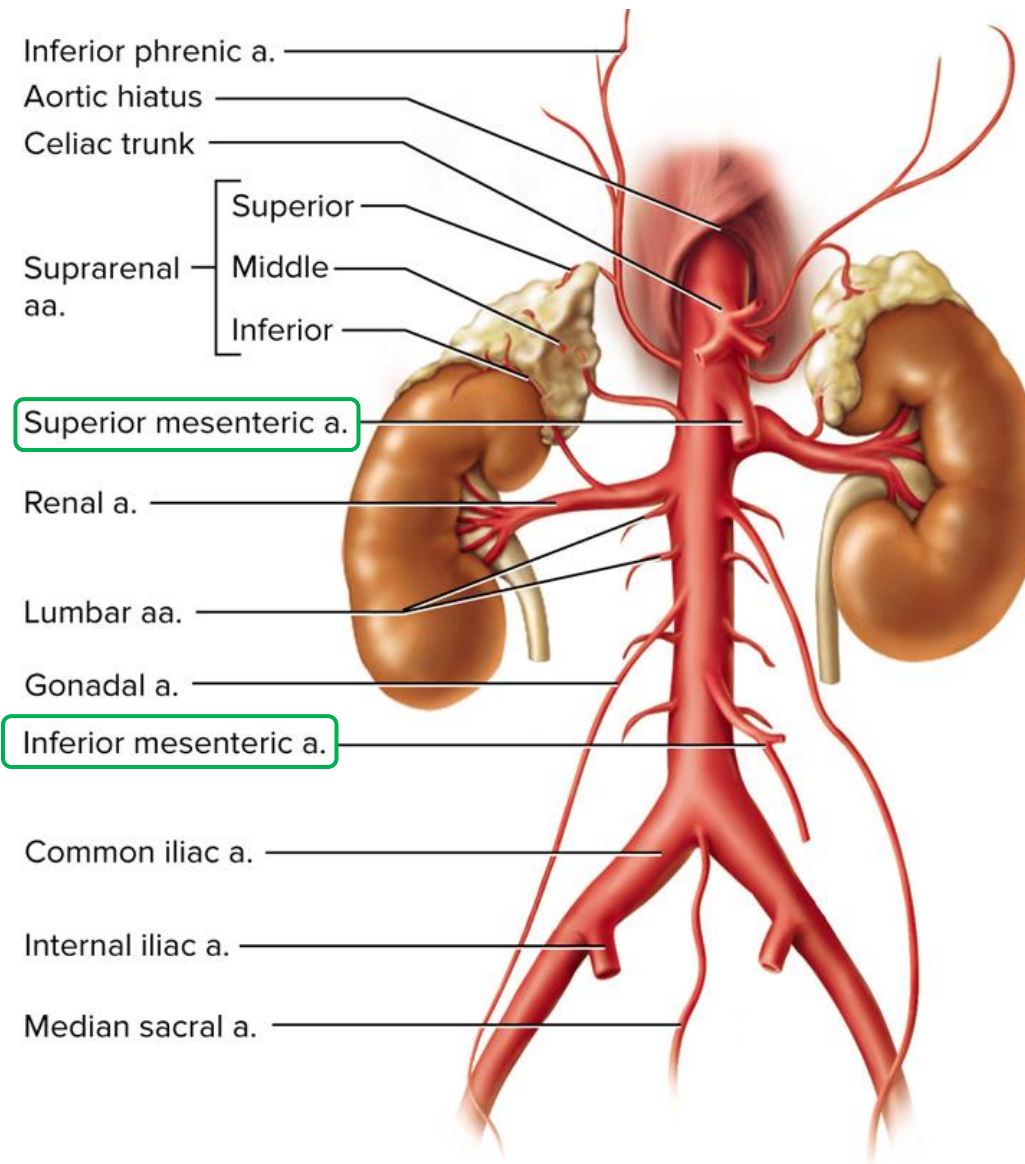


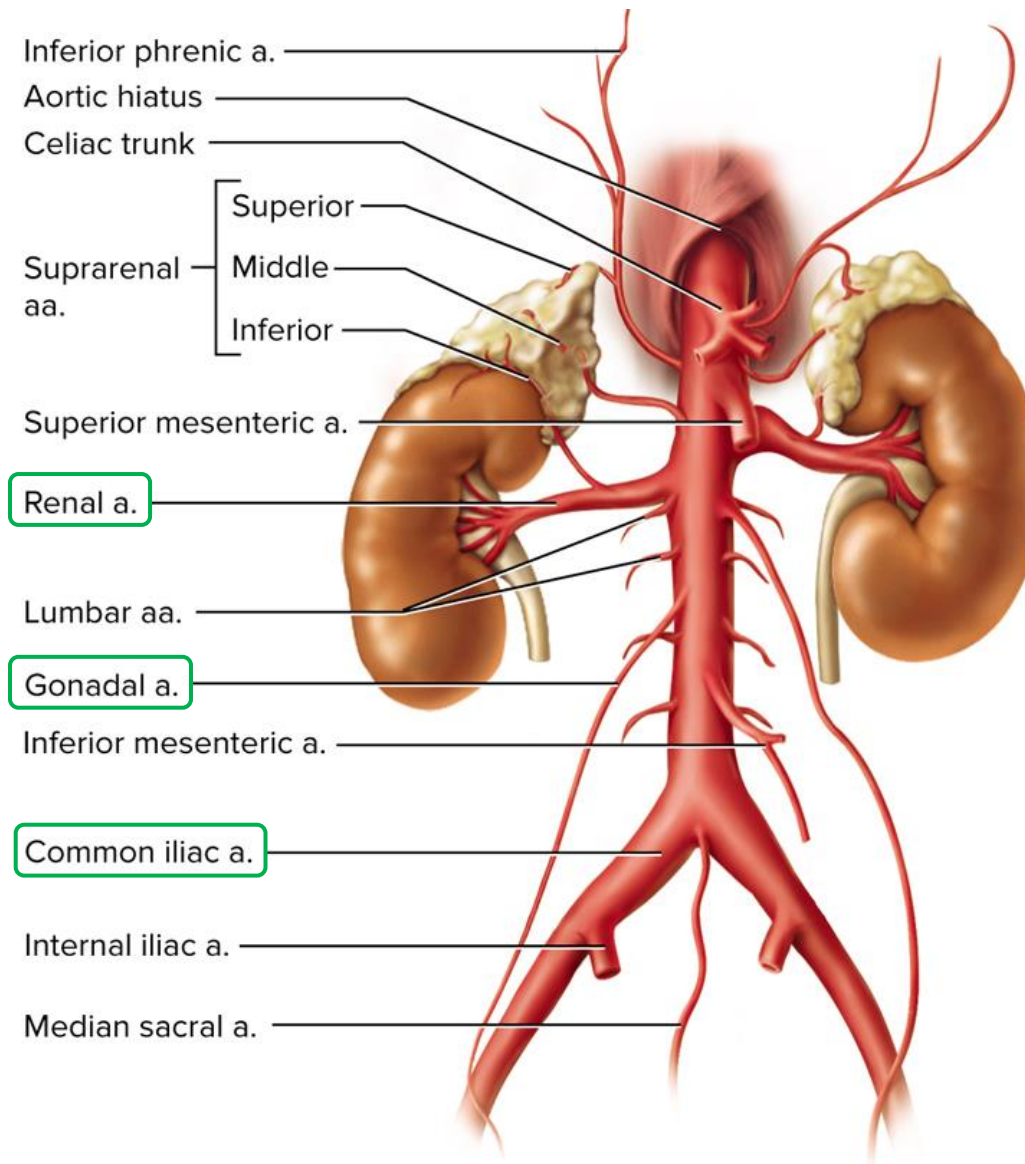


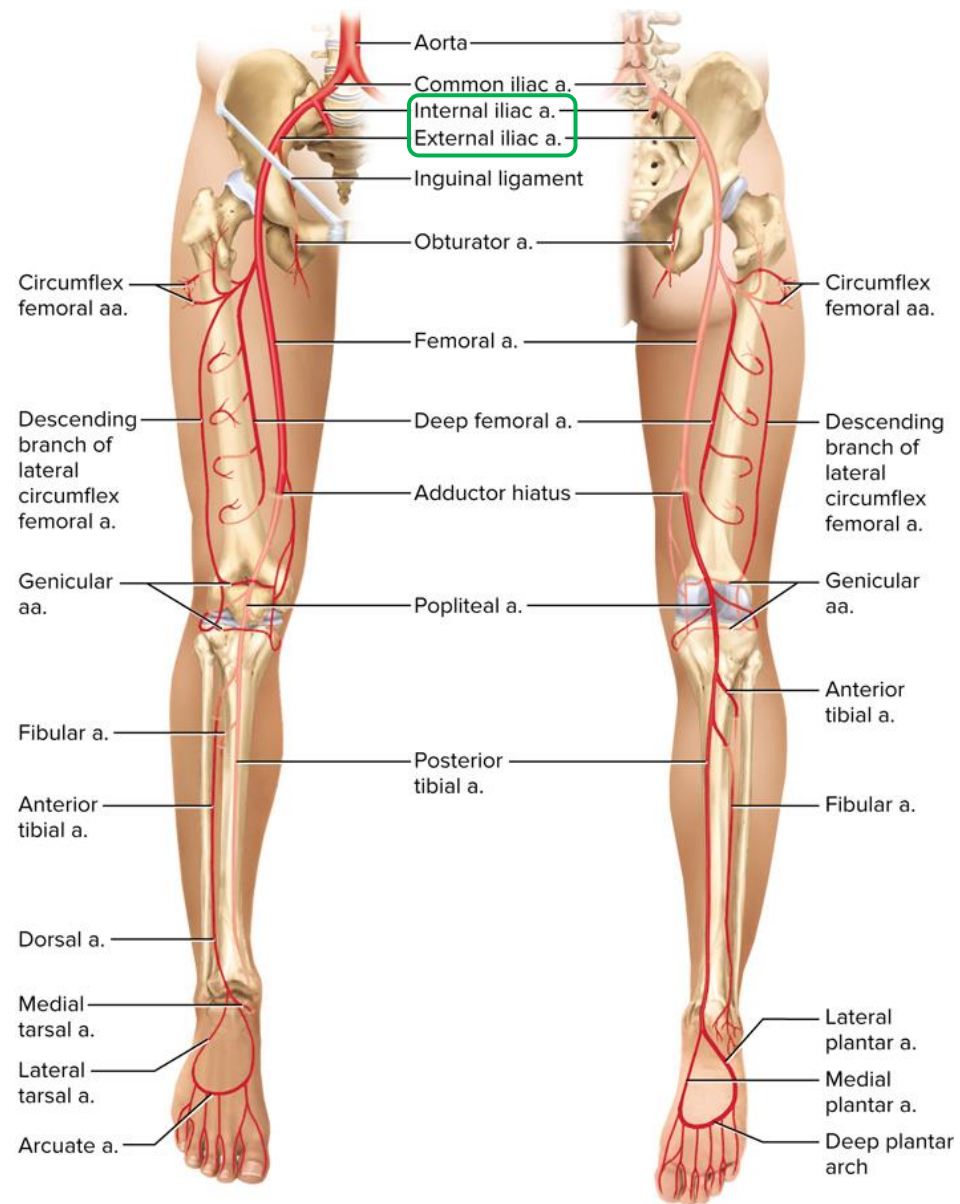


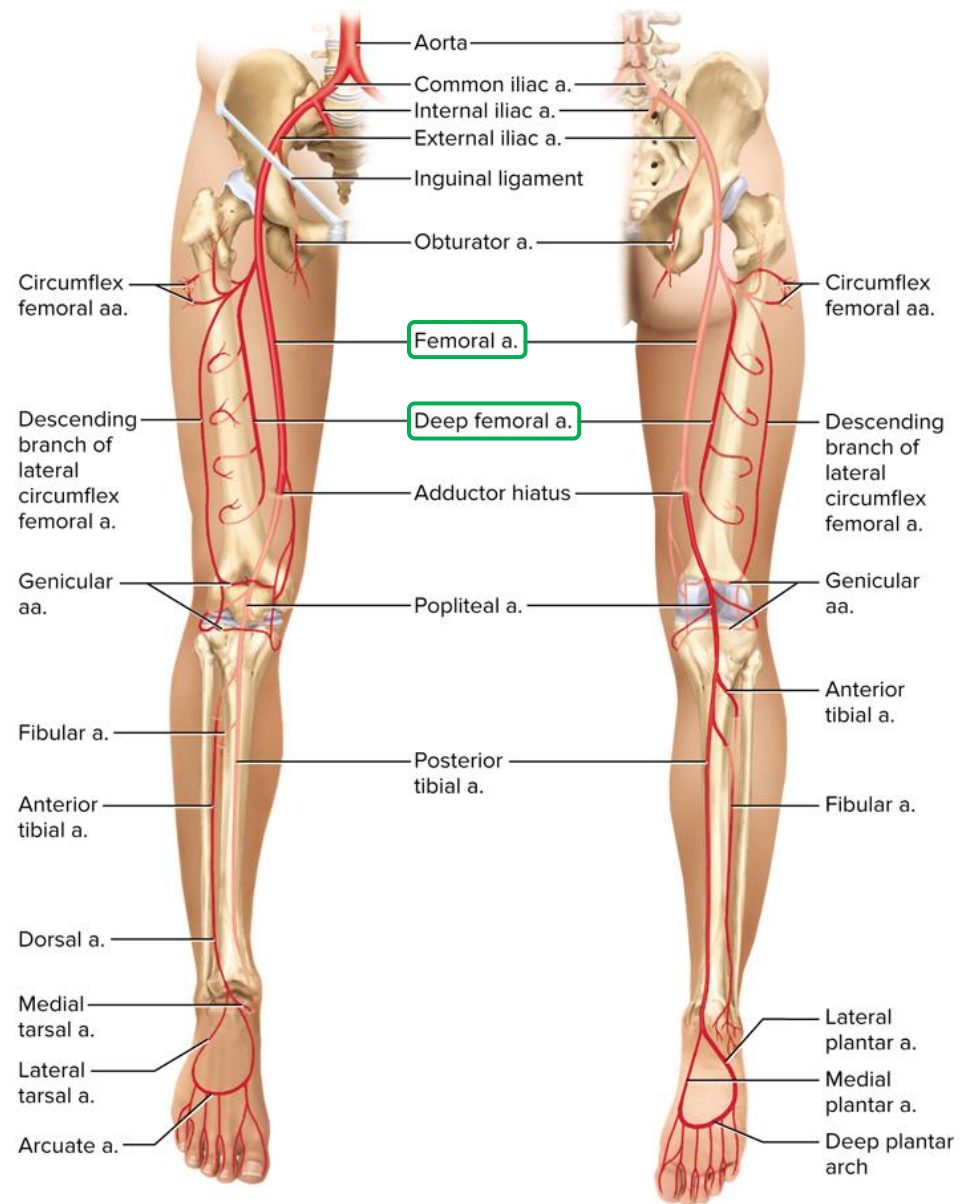


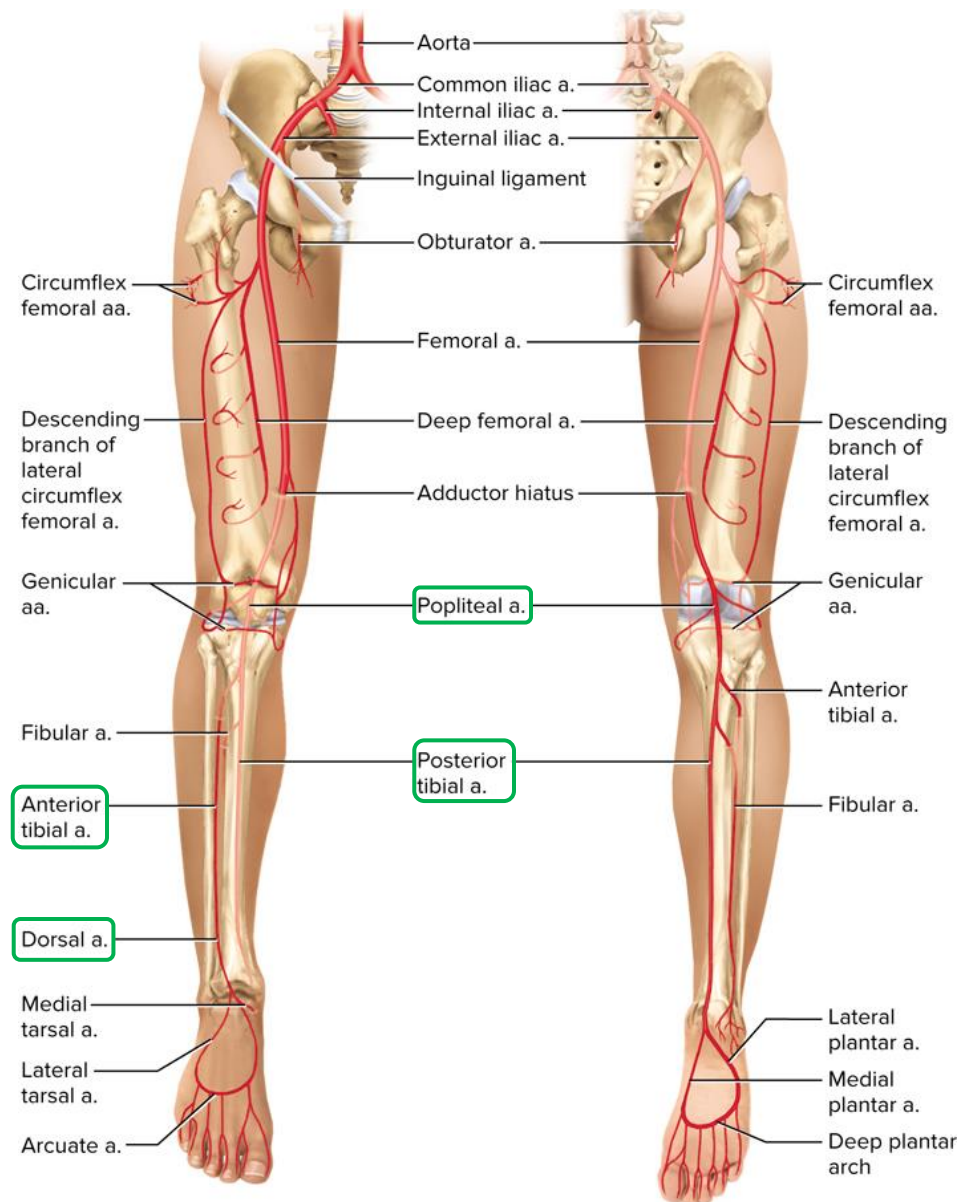


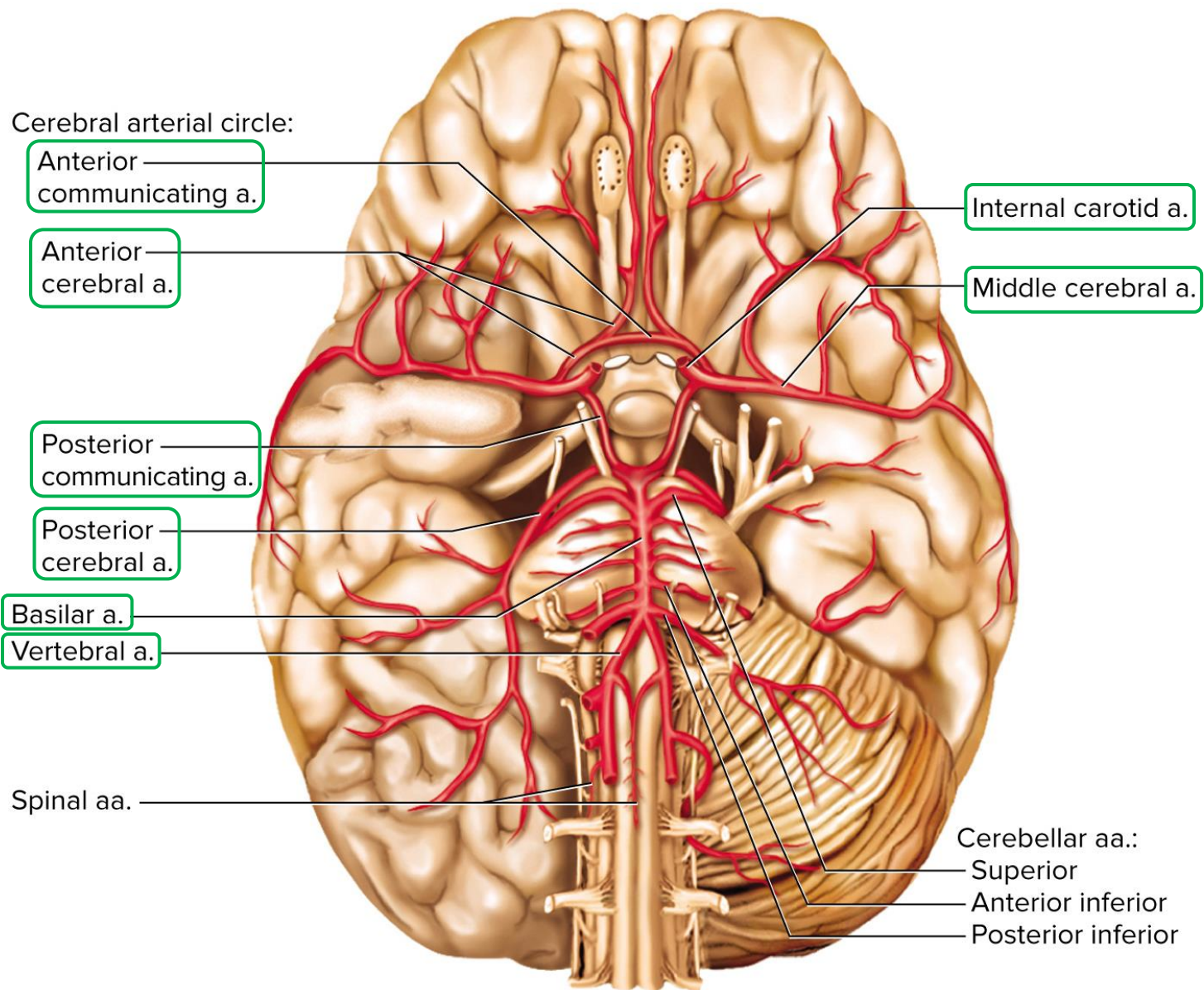


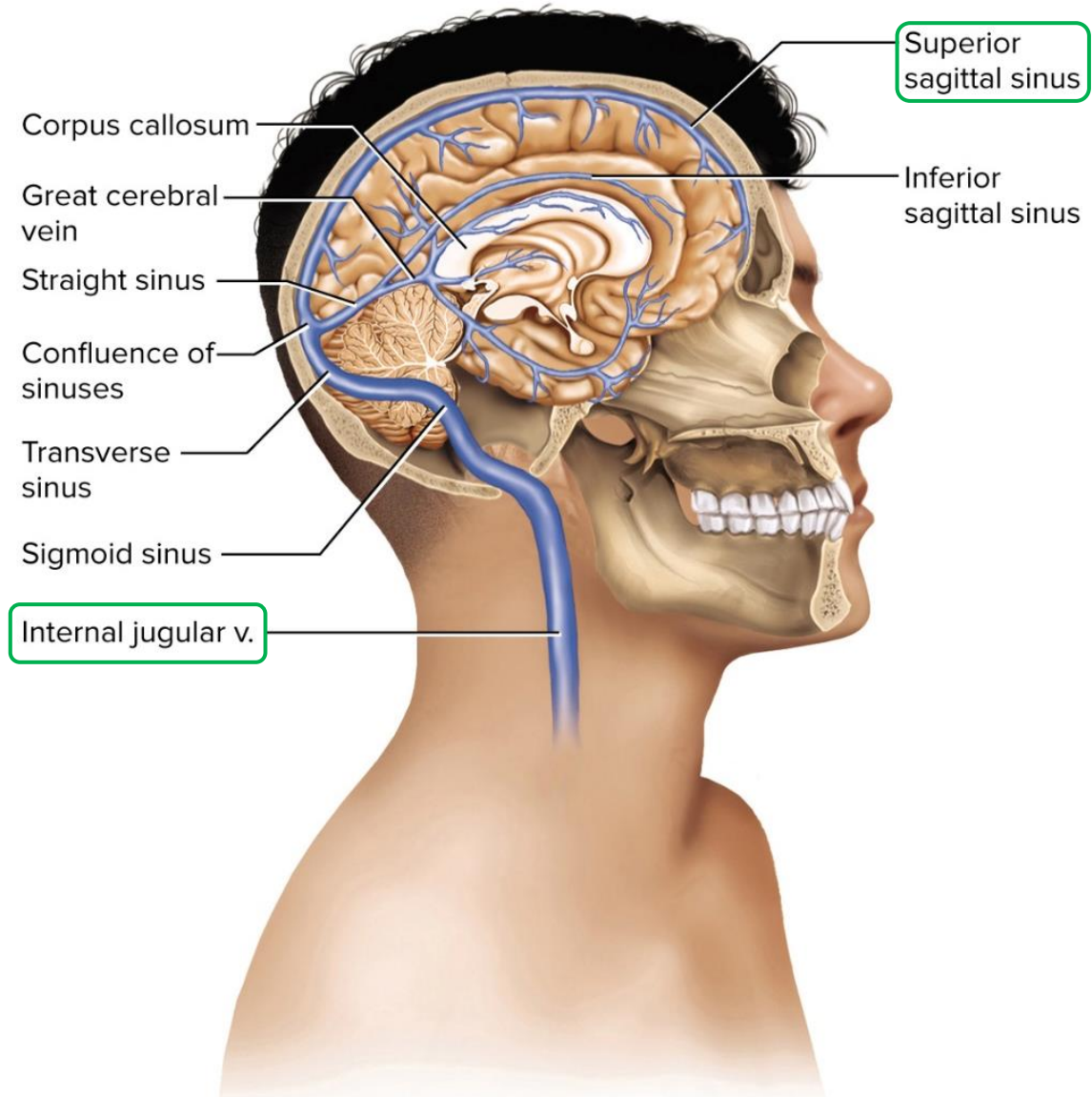


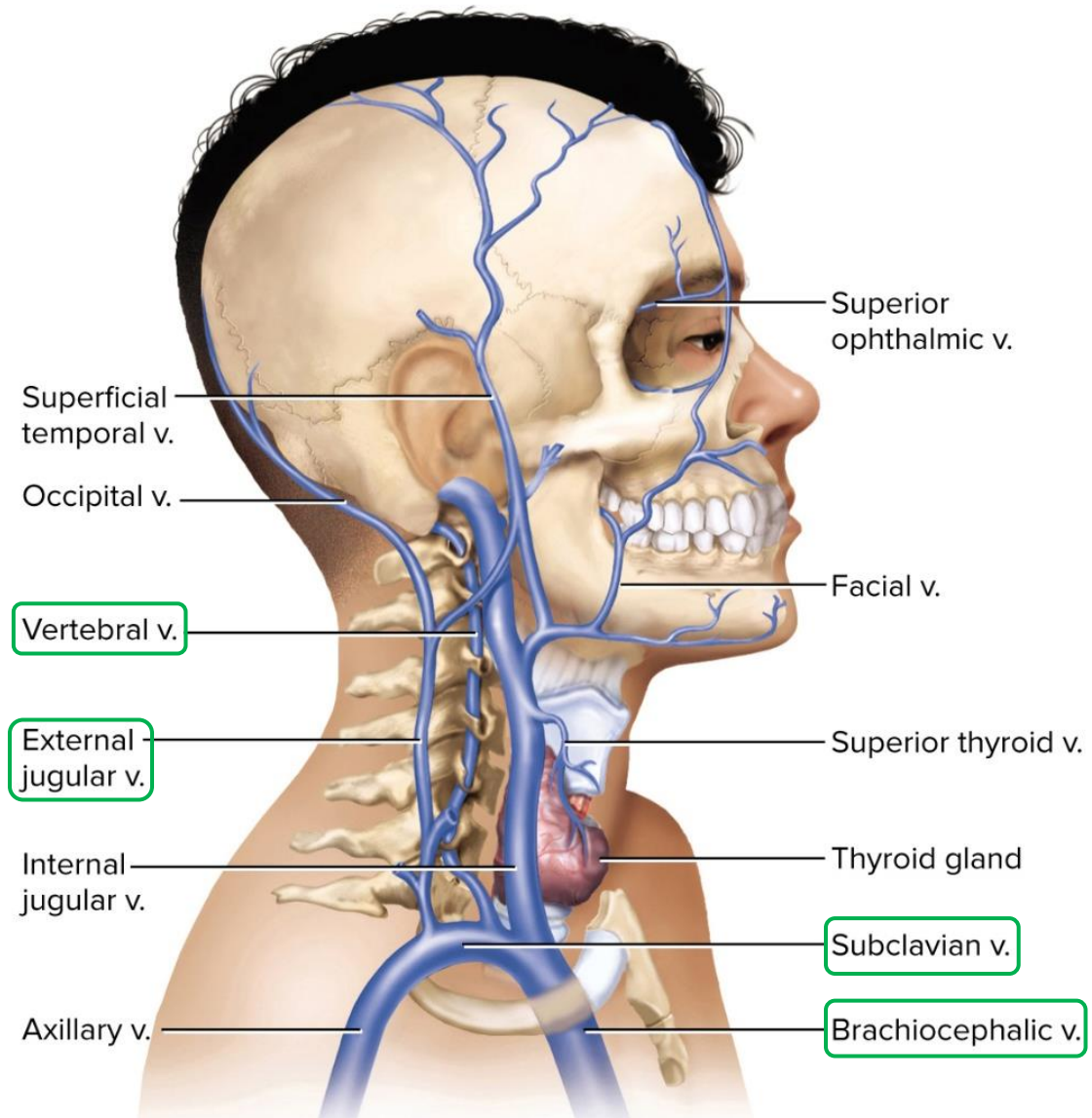


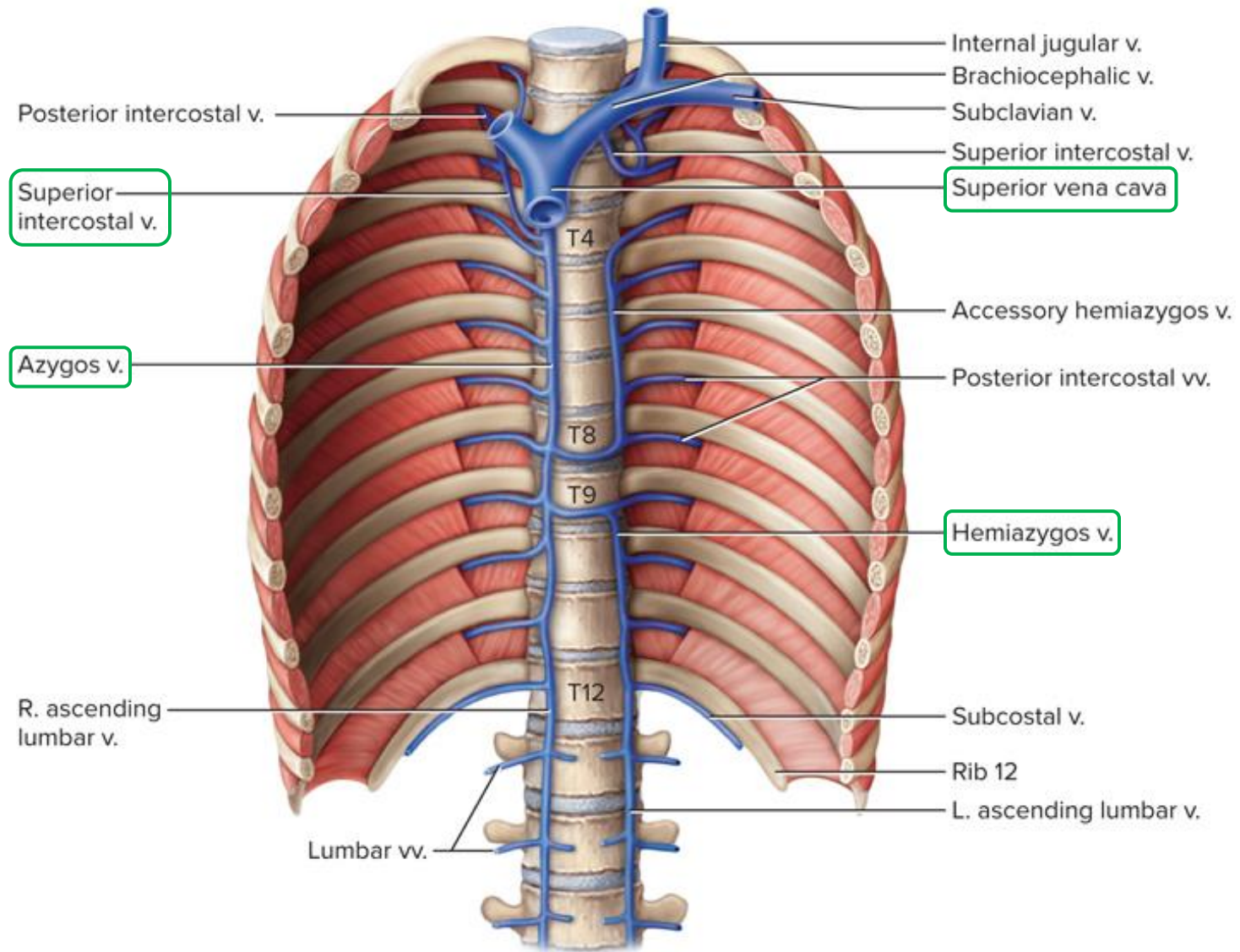


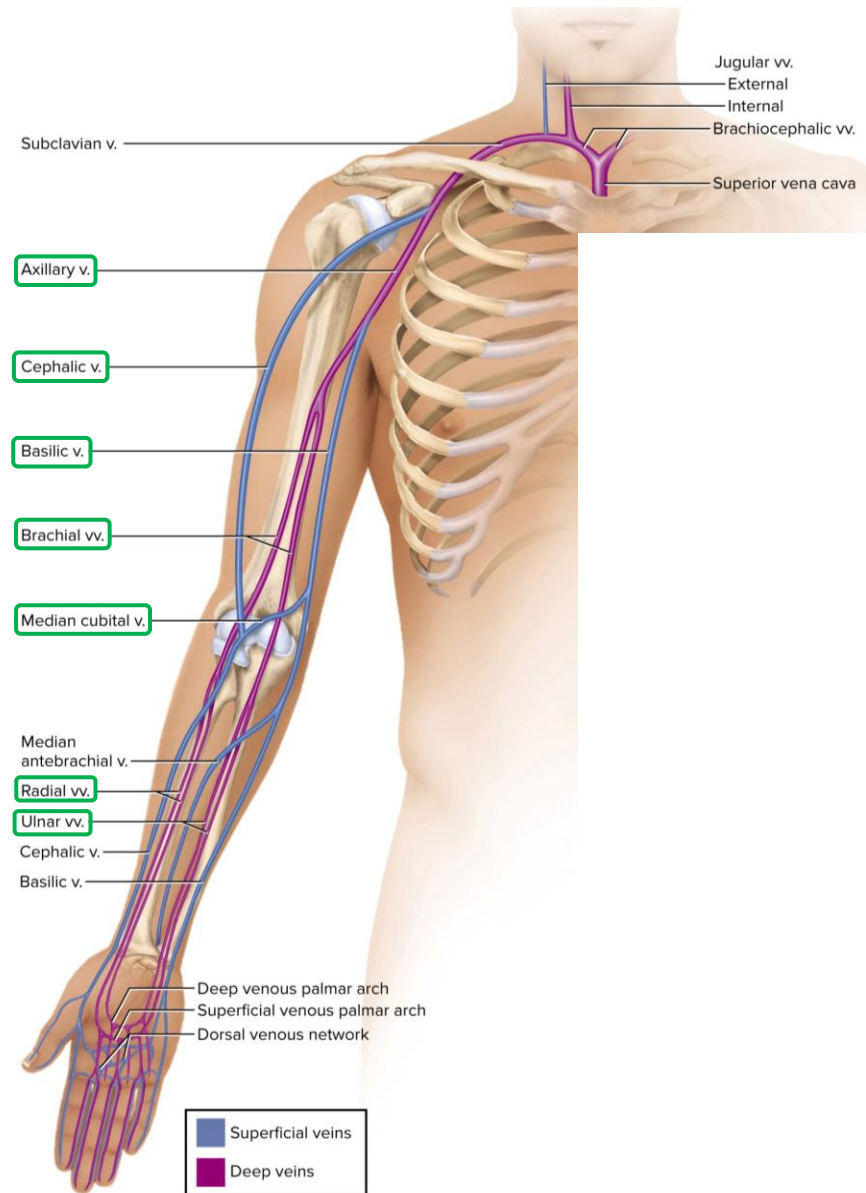


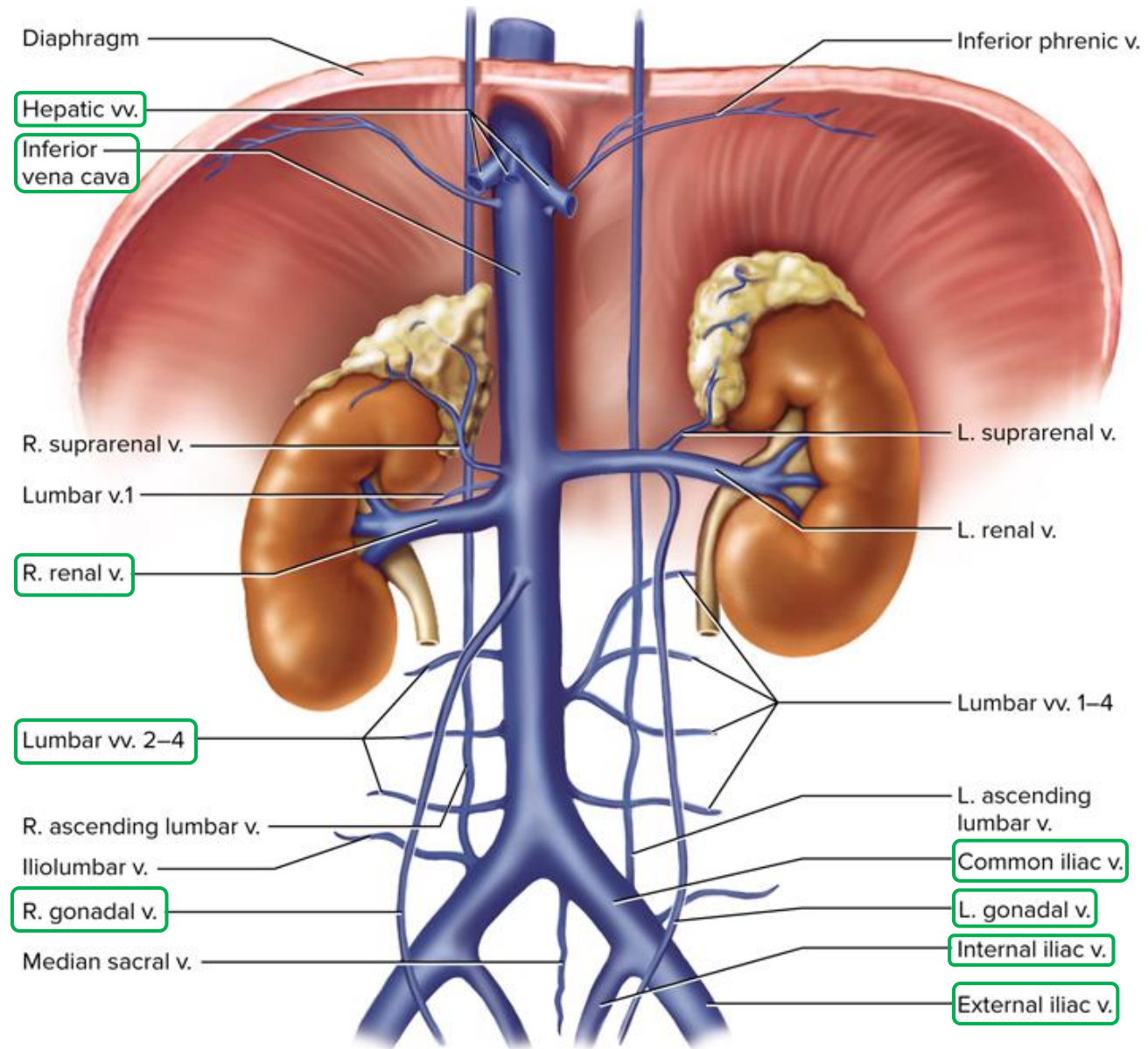


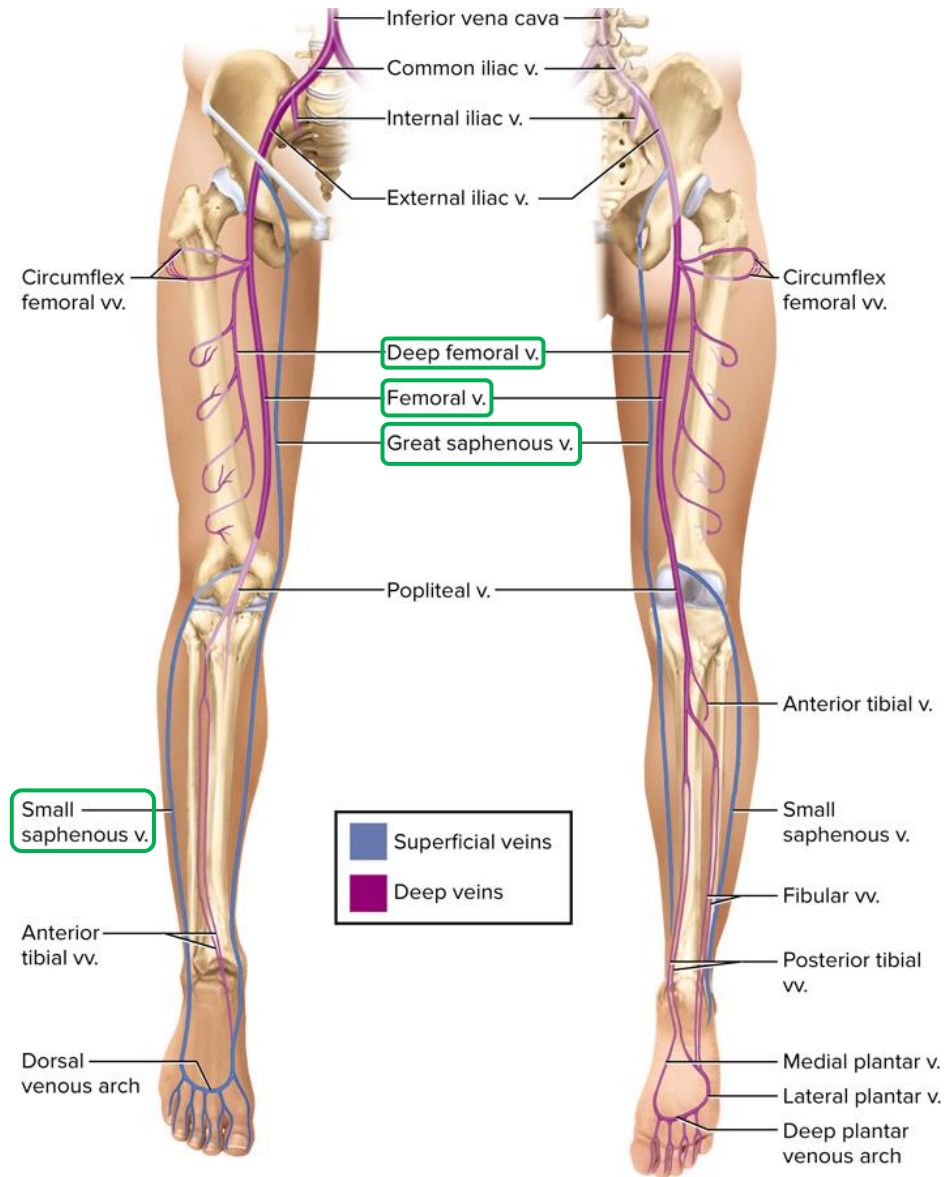


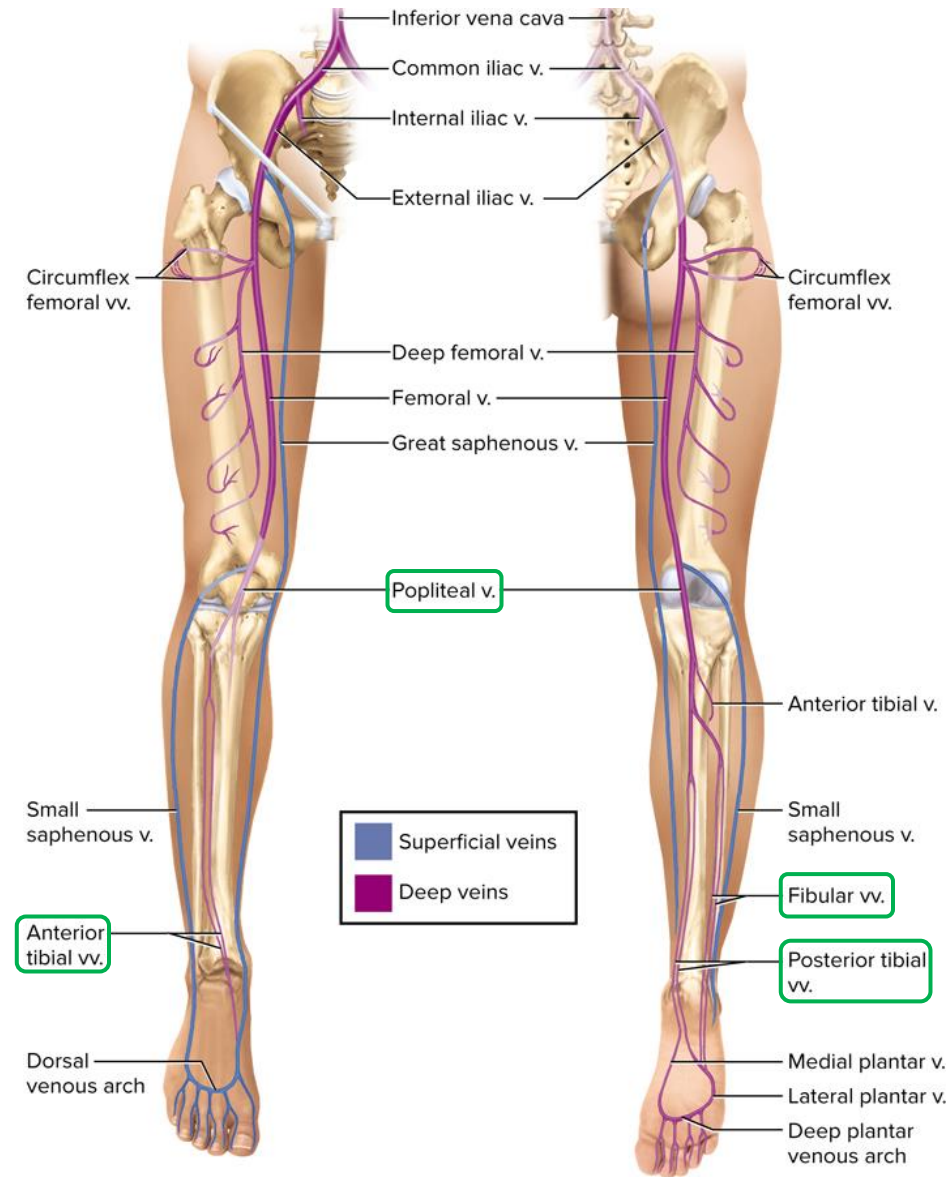


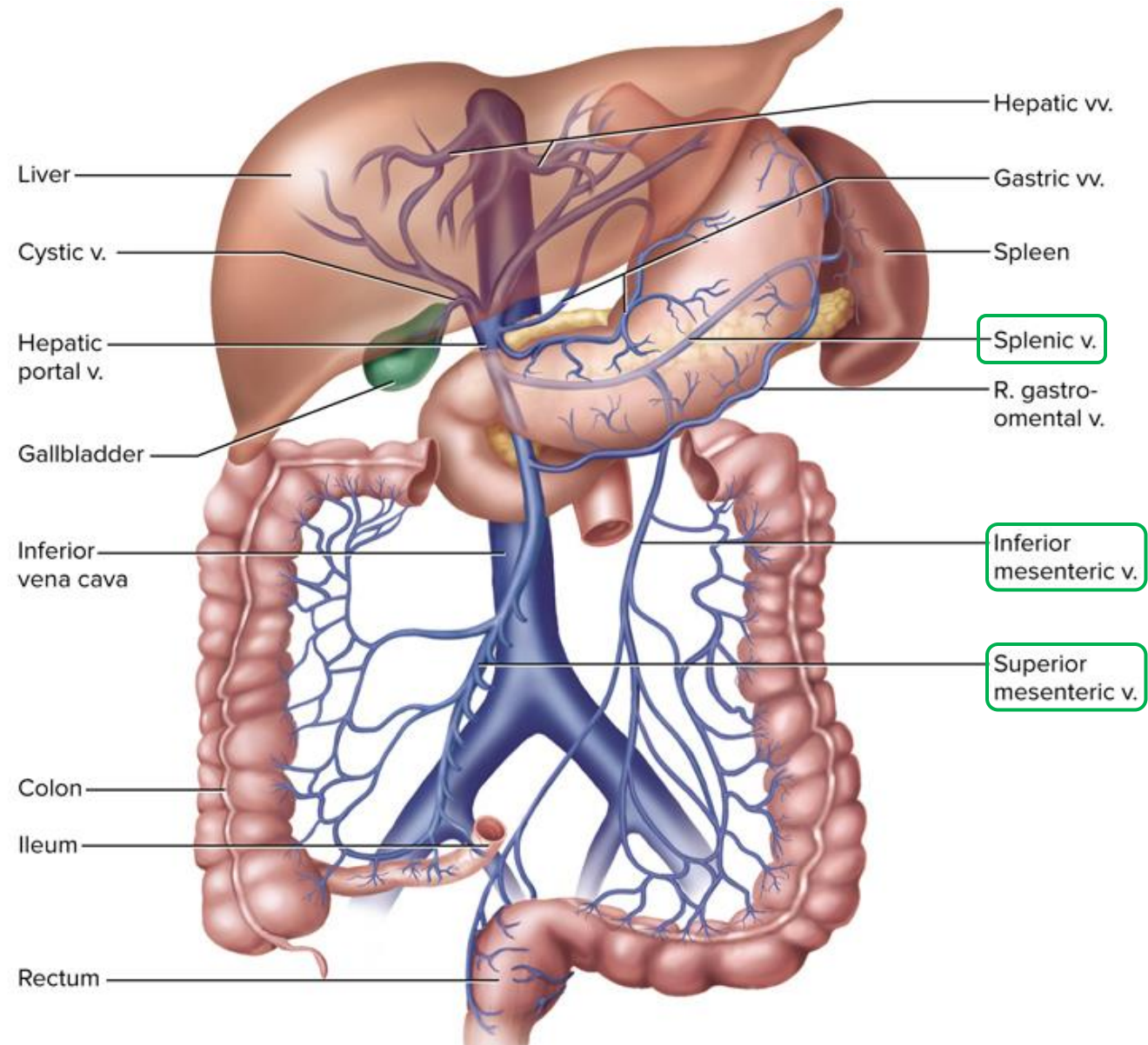




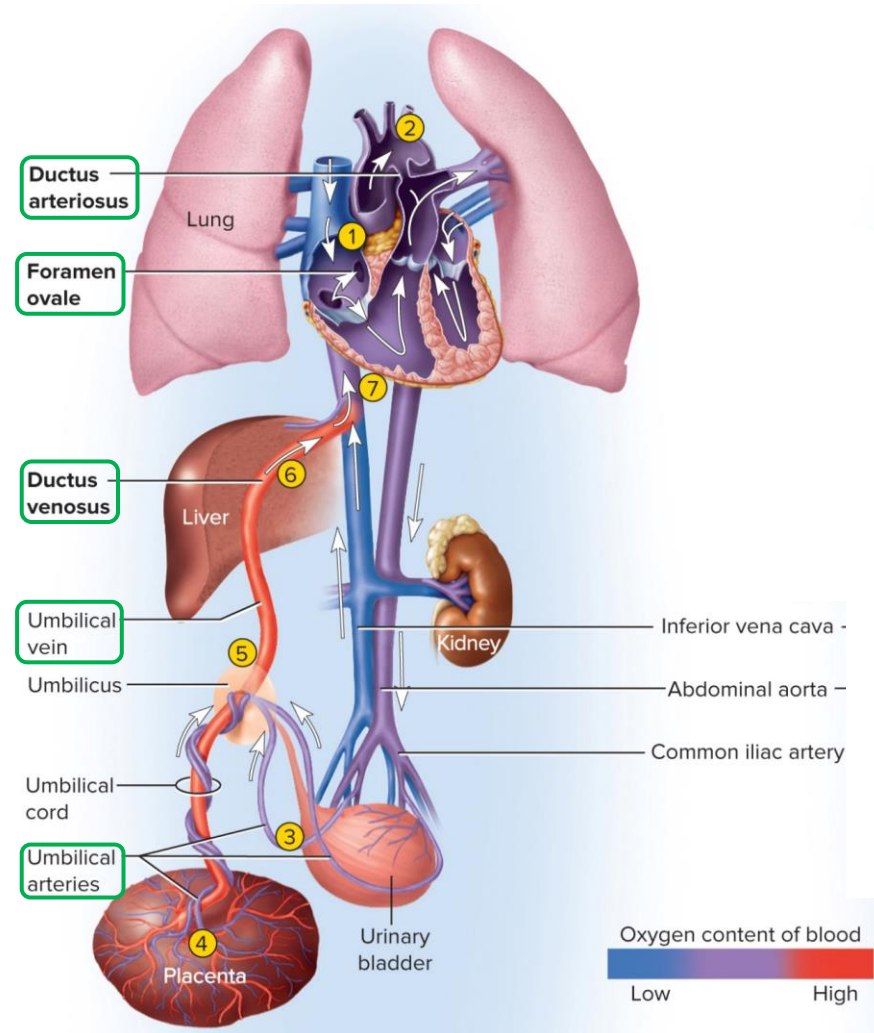








Fetal Circulation



Fetal circulation

- 1 Blood bypasses the lungs by flowing directly from the right atrium through the foramen ovale into the left atrium.
- 2 Blood also bypasses the lungs by flowing from the pulmonary trunk through the ductus arteriosus into the aorta.
- 3 Oxygen-poor, waste-laden blood flows through two umbilical arteries to the placenta.
- 4 The placenta disposes of CO₂ and other wastes and reoxygenates the blood.
- 5 Oxygenated blood returns to the fetus through the umbilical vein.
- 6 Placental blood bypasses the liver by flowing through the ductus venosus into the inferior vena cava (IVC).
- 7 Placental blood from the umbilical vein mixes with fetal blood from the IVC and returns to the heart.